



MKD

2026-07-09

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- 2 Mathematics
- 3 Data structures
- 4 Numerical
- 5 Number theory
- 6 Combinatorial
- 7 Graph
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Contest (1)

dbg.h20 lines

```
#define dbg(...) cerr<<"DEBUG:["<<#__VA_ARGS__<<"] = [" ,\
_print(__VA_ARGS__),cerr<<"]\n"
template<class T, class=void>struct cntr : false_type {};
template<class T>struct cntr<T,
void_t<decltype(declval<T>()).begin()),
decltype(declval<T>()).end()),
typename T::value_type> : true_type {};
template<class T>using raw=remove_cv_t<remove_reference_t<T>>;
template<class T>enable_if_t!cntr<T>::value>
_print(const T &x) { cerr << x; }
template<class T>enable_if_t<cntr<T>::value>_print(const T &v){
cerr << "["; bool f = false;
for (const auto &x: v)
f ? cerr << (cntr<raw<decltype(x)>>::value ?",\n":", ,"):
cerr, f = true, _print(x);
cerr << "]" ;
}
template<class T, class ... A>
void _print(const T &x, const A &... a) {
_print(x);(cerr << " , ", _print(a)), ... );}
```

template.cpp25 lines

```
#include <bits/stdc++.h>
using namespace std;
// #define LOCAL_DEBUG
#ifdef LOCAL_DEBUG
#include "dbg.h"
#else
#define dbg(...) 42
#endif
#define rep(i, a, b) for(int i = a; i < (b); ++i)
#define all(x) begin(x), end(x)
#define sz(x) (int)(x).size()
using ll = long long;
using pii = pair<int, int>;
using vi = vector<int>;
void init() {}
```

```
void solve() {}
signed main() {
init();
cin.tie(nullptr), ios::sync_with_stdio(false);
// cin.exceptions(cin.failbit);
int T = 1;
// cin >> T;
while (T--)solve();
return 0;
}
```

1.1 checker

cek.sh19 lines

```
#!/usr/bin/env bash
set -u
cd "$(dirname "$0")"
g++ gen.cpp -o gen -std=c++17 -O2
g++ std.cpp -o std -std=c++17 -O2
g++ sol.cpp -o sol -std=c++17 -O2
cnt=0
while true; do
cnt=$((cnt + 1))
echo "Testing case #$cnt"
./gen "${1-}" > in.txt
./std < in.txt > out1.txt
./sol < in.txt > out2.txt
if ! diff -q out1.txt out2.txt > /dev/null; then
echo "WA on case #$cnt"
diff out1.txt out2.txt
read -r -p "Press Enter to continue..."
fi
done
```

cek.bat19 lines

```
@echo off
cd "%~dp0"
g++ gen.cpp -o gen -std=c++17 -O2
g++ std.cpp -o std -std=c++17 -O2
g++ sol.cpp -o sol -std=c++17 -O2
set cnt=0
:loop
set /a cnt+=1
echo Testing case %cnt%
gen %1 > in.txt
std < in.txt > out1.txt
sol < in.txt > out2.txt
fc out1.txt out2.txt > nul
if errorlevel 1 (
echo WA on case %cnt%
fc out1.txt out2.txt
pause
)
goto loop
```

gen.h22 lines

```
uint64_t seed = chrono::steady_clock::now()
.time_since_epoch().count() ^ random_device{}();
mt19937 rng(seed);
template<class T>
T randi(T l,T r){return uniform_int_distribution<T>(l,r)(rng);}
vi randvi(int n, int l, int r) {
vi v(n);
rep(i, 0, sz(v))v[i] = randi(l, r);
return v;
}
vi randp(int n) {
```

```
vi p(n);
iota(all(p), 1);
shuffle(all(p), rng);
return p;
}
int main(int argc, char *argv[]) {
if (argc >= 2)
seed = stoull(argv[1]), rng.seed(seed);
cerr << seed << endl;
return 0;
}
```

.bashrc3 lines

```
alias c='g++ -Wall -Wconversion -Wfatal-errors -g -std=c++17 \
-fsanitize=undefined,address'
xmodmap -e 'clear lock' -e 'keycode 66=less greater' #caps = <>
```

.vimrc6 lines

```
set cin aw ai is ts=4 sw=4 tm=50 nu noeb bg=dark ru cul
sy on | im jk <esc> | im kj <esc> | no ; :
" Select region and then type :Hash to hash your selection.
" Useful for verifying that there aren't mistypes.
ca Hash w !cpp -dD -P -fpreprocessed \ | tr -d '[:space:]' \
\ | md5sum \ | cut -c-6
```

hash.sh3 lines

```
# Hashes a file, ignoring all whitespace and comments. Use for
# verifying that code was correctly typed.
cpp -dD -P -fpreprocessed | tr -d '[:space:]' | md5sum |cut -c-6
```

troubleshoot.txt52 lines

```
Pre-submit:
Write a few simple test cases if sample is not enough.
Are time limits close? If so, generate max cases.
Is the memory usage fine?
Could anything overflow?
Make sure to submit the right file.
```

```
Wrong answer:
Print your solution! Print debug output, as well.
Are you clearing all data structures between test cases?
Can your algorithm handle the whole range of input?
Read the full problem statement again.
Do you handle all corner cases correctly?
Have you understood the problem correctly?
Any uninitialized variables?
Any overflows?
Confusing N and M, i and j, etc.?
Are you sure your algorithm works?
What special cases have you not thought of?
Are you sure the STL functions you use work as you think?
Add some assertions, maybe resubmit.
Create some testcases to run your algorithm on.
Go through the algorithm for a simple case.
Go through this list again.
Explain your algorithm to a teammate.
Ask the teammate to look at your code.
Go for a small walk, e.g. to the toilet.
Is your output format correct? (including whitespace)
Rewrite your solution from the start or let a teammate do it.
```

```
Runtime error:
Have you tested all corner cases locally?
Any uninitialized variables?
Are you reading or writing outside the range of any vector?
```

Any assertions that might fail?
 Any possible division by 0? (mod 0 for example)
 Any possible infinite recursion?
 Invalidated pointers or iterators?
 Are you using too much memory?
 Debug with resubmits (e.g. remapped signals, see Various).

Time limit exceeded:
 Do you have any possible infinite loops?
 What is the complexity of your algorithm?
 Are you copying a lot of unnecessary data? (References)
 How big is the input and output? (consider scanf)
 Avoid vector, map. (use arrays/unordered_map)
 What do your teammates think about your algorithm?

Memory limit exceeded:
 What is the max amount of memory your algorithm should need?
 Are you clearing all data structures between test cases?

Mathematics (2)

2.1 Equations

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The extremum is given by $x = -\frac{b}{2a}$.

$$\begin{aligned} ax + by &= e & x &= \frac{ed - bf}{ad - bc} \\ cx + dy &= f & y &= \frac{af - ec}{ad - bc} \end{aligned}$$

In general, given an equation $Ax = b$, the solution to a variable x_i is given by

$$x_i = \frac{\det A'_i}{\det A}$$

where A'_i is A with the i 'th column replaced by b .

2.2 Recurrences

If $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$, and $r_1, \dots, r_k$ are distinct roots of $x^k - c_1 x^{k-1} - \dots - c_k$, there are $d_1, \dots, d_k$ s.t.

$$a_n = d_1 r_1^n + \dots + d_k r_k^n.$$

Non-distinct roots r become polynomial factors, e.g.

$$a_n = (d_1 n + d_2) r^n.$$

2.3 Trigonometry

$$\sin(v + w) = \sin v \cos w + \cos v \sin w$$

$$\cos(v + w) = \cos v \cos w - \sin v \sin w$$

$$\tan(v + w) = \frac{\tan v + \tan w}{1 - \tan v \tan w}$$

$$\sin v + \sin w = 2 \sin \frac{v + w}{2} \cos \frac{v - w}{2}$$

$$\cos v + \cos w = 2 \cos \frac{v + w}{2} \cos \frac{v - w}{2}$$

$$(V + W) \tan(v - w)/2 = (V - W) \tan(v + w)/2$$

where V, W are lengths of sides opposite angles v, w .

$$a \cos x + b \sin x = r \cos(x - \phi)$$

$$a \sin x + b \cos x = r \sin(x + \phi)$$

where $r = \sqrt{a^2 + b^2}$, $\phi = \text{atan2}(b, a)$.

2.4 Geometry

2.4.1 Triangles

Side lengths: a, b, c

$$\text{Semiperimeter: } p = \frac{a + b + c}{2}$$

$$\text{Area: } A = \sqrt{p(p - a)(p - b)(p - c)}$$

$$\text{Circumradius: } R = \frac{abc}{4A}$$

$$\text{Inradius: } r = \frac{A}{p}$$

Length of median (divides triangle into two equal-area triangles):

$$m_a = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2}$$

Length of bisector (divides angles in two):

$$s_a = \sqrt{bc \left[1 - \left(\frac{a}{b + c} \right)^2 \right]}$$

$$\text{Law of sines: } \frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c} = \frac{1}{2R}$$

$$\text{Law of cosines: } a^2 = b^2 + c^2 - 2bc \cos \alpha$$

$$\text{Law of tangents: } \frac{a + b}{a - b} = \frac{\tan \frac{\alpha + \beta}{2}}{\tan \frac{\alpha - \beta}{2}}$$

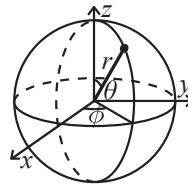
2.4.2 Quadrilaterals

With side lengths a, b, c, d , diagonals e, f , diagonals angle θ , area A and magic flux $F = b^2 + d^2 - a^2 - c^2$:

$$4A = 2ef \cdot \sin \theta = F \tan \theta = \sqrt{4e^2 f^2 - F^2}$$

For cyclic quadrilaterals the sum of opposite angles is 180° ,
 $2ef = ac + bd = (p - a)(p - b)(p - c)(p - d)$.

2.4.3 Spherical coordinates



$$\begin{aligned} x &= r \sin \theta \cos \phi & r &= \sqrt{x^2 + y^2 + z^2} \\ y &= r \sin \theta \sin \phi & \theta &= \arccos(z / \sqrt{x^2 + y^2 + z^2}) \\ z &= r \cos \theta & \phi &= \text{atan2}(y, x) \end{aligned}$$

2.5 Derivatives/Integrals

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1 - x^2}} \quad \frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1 - x^2}}$$

$$\frac{d}{dx} \tan x = 1 + \tan^2 x \quad \frac{d}{dx} \arctan x = \frac{1}{1 + x^2}$$

$$\int \tan ax = -\frac{\ln |\cos ax|}{a} \quad \int x \sin ax = \frac{\sin ax - ax \cos ax}{a^2}$$

$$\int e^{-x^2} = \frac{\sqrt{\pi}}{2} \text{erf}(x) \quad \int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1)$$

Integration by parts:

$$\int_a^b f(x)g(x)dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x)dx$$

2.6 Sums

$$c^a + c^{a+1} + \dots + c^b = \frac{c^{b+1} - c^a}{c - 1}, c \neq 1$$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(2n+1)(n+1)}{6}$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

2.7 Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, (-\infty < x < \infty)$$

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, (-1 < x \leq 1)$$

$$\sqrt{1 + x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \dots, (-1 \leq x \leq 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, (-\infty < x < \infty)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, (-\infty < x < \infty)$$

2.8 Probability theory

Let X be a discrete random variable with probability $p_X(x)$ of assuming the value x . It will then have an expected value (mean) $\mu = \mathbb{E}(X) = \sum_x x p_X(x)$ and variance $\sigma^2 = V(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_x (x - \mathbb{E}(X))^2 p_X(x)$ where σ is the standard deviation. If X is instead continuous it will have a probability density function $f_X(x)$ and the sums above will instead be integrals with $p_X(x)$ replaced by $f_X(x)$.

Expectation is linear:

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

For independent X and Y ,

$$V(aX + bY) = a^2V(X) + b^2V(Y).$$

2.8.1 Discrete distributions

Binomial distribution

The number of successes in n independent yes/no experiments, each which yields success with probability p is

$\text{Bin}(n, p)$, $n = 1, 2, \dots$, $0 \leq p \leq 1$.

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mu = np, \sigma^2 = np(1-p)$$

$\text{Bin}(n, p)$ is approximately $\text{Po}(np)$ for small p .

First success distribution

The number of trials needed to get the first success in independent yes/no experiments, each which yields success with probability p is $\text{Fs}(p)$, $0 \leq p \leq 1$.

$$p(k) = p(1-p)^{k-1}, k = 1, 2, \dots$$

$$\mu = \frac{1}{p}, \sigma^2 = \frac{1-p}{p^2}$$

Poisson distribution

The number of events occurring in a fixed period of time t if these events occur with a known average rate κ and independently of the time since the last event is $\text{Po}(\lambda)$, $\lambda = t\kappa$.

$$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, k = 0, 1, 2, \dots$$

$$\mu = \lambda, \sigma^2 = \lambda$$

2.8.2 Continuous distributions

Uniform distribution

If the probability density function is constant between a and b and 0 elsewhere it is $\text{U}(a, b)$, $a < b$.

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

$$\mu = \frac{a+b}{2}, \sigma^2 = \frac{(b-a)^2}{12}$$

Exponential distribution

The time between events in a Poisson process is $\text{Exp}(\lambda)$, $\lambda > 0$.

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\mu = \frac{1}{\lambda}, \sigma^2 = \frac{1}{\lambda^2}$$

UnionFind RMQ FenwickTree SubMatrix

Normal distribution

Most real random values with mean μ and variance σ^2 are well described by $\mathcal{N}(\mu, \sigma^2)$, $\sigma > 0$.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ then

$$aX_1 + bX_2 + c \sim \mathcal{N}(\mu_1 + \mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2)$$

2.9 Markov chains

A *Markov chain* is a discrete random process with the property that the next state depends only on the current state. Let $X_1, X_2, \dots$ be a sequence of random variables generated by the Markov process. Then there is a transition matrix $\mathbf{P} = (p_{ij})$, with $p_{ij} = \Pr(X_n = i | X_{n-1} = j)$, and $\mathbf{p}^{(n)} = \mathbf{P}^n \mathbf{p}^{(0)}$ is the probability distribution for X_n (i.e., $p_i^{(n)} = \Pr(X_n = i)$), where $\mathbf{p}^{(0)}$ is the initial distribution.

π is a stationary distribution if $\pi = \pi\mathbf{P}$. If the Markov chain is *irreducible* (it is possible to get to any state from any state), then $\pi_i = \frac{1}{\mathbb{E}(T_i)}$ where $\mathbb{E}(T_i)$ is the expected time between two visits in state i . π_j / π_i is the expected number of visits in state j between two visits in state i .

For a connected, undirected and non-bipartite graph, where the transition probability is uniform among all neighbors, π_i is proportional to node i 's degree.

A Markov chain is *ergodic* if the asymptotic distribution is independent of the initial distribution. A finite Markov chain is ergodic iff it is irreducible and *aperiodic* (i.e., the gcd of cycle lengths is 1). $\lim_{k \rightarrow \infty} \mathbf{P}^k = 1\pi$.

A Markov chain is an A-chain if the states can be partitioned into two sets $\mathbf{A}$ and $\mathbf{G}$, such that all states in $\mathbf{A}$ are absorbing ($p_{ii} = 1$), and all states in $\mathbf{G}$ leads to an absorbing state in $\mathbf{A}$. The probability for absorption in state $i \in \mathbf{A}$, when the initial state is j , is $a_{ij} = p_{ij} + \sum_{k \in \mathbf{G}} a_{ik} p_{kj}$. The expected time until absorption, when the initial state is i , is $t_i = 1 + \sum_{k \in \mathbf{G}} p_{ki} t_k$.

Data structures (3)

UnionFind.h

Description: 并查集数据结构。

Time: $\mathcal{O}(\alpha(N))$

7aa27c, 14 lines

```
struct UF {
    vi e;
    UF(int n) : e(n, -1) {}
    bool sameSet(int a, int b) { return find(a) == find(b); }
    int size(int x) { return -e[find(x)]; }
    int find(int x) { return e[x] < 0 ? x : e[x] = find(e[x]); }
    bool join(int a, int b) {
        a = find(a), b = find(b);
        if (a == b) return false;
        if (e[a] > e[b]) swap(a, b);
```

```
    e[a] += e[b]; e[b] = a;
    return true;
}
};
```

RMQ.h

Description: 区间最小值查询。返回 $\min(V[a], V[a+1], \dots, V[b-1])$ 。

Usage: `RMQ rmq(values);`

`rmq.query(inclusive, exclusive);`

Time: $\mathcal{O}(|V| \log |V| + Q)$

510c32, 15 lines

```
template<class T>struct RMQ {
    vector<vector<T>> jmp;
    RMQ(const vector<T>& V) : jmp(1, V) { // pw = 2^k-1
        for (int pw = 1, k = 1; pw * 2 <= sz(V); pw *= 2, ++k) {
            jmp.emplace_back(sz(V) - pw * 2 + 1);
            rep(j, 0, sz(jmp[k]))
                jmp[k][j] = min(jmp[k-1][j], jmp[k-1][j+pw]);
        }
    }
    T query(int a, int b) {
        assert(a < b); // or return inf if a == b
        int dep = 31 - __builtin_clz(b - a);
        return min(jmp[dep][a], jmp[dep][b - (1 << dep)]);
    }
};
```

FenwickTree.h

Description: 树状数组。计算前缀和 $a[0] + a[1] + \dots + a[\text{pos} - 1]$, 并通过计算新旧值之差更新单个元素 $a[i]$ 。

Time: Both operations are $\mathcal{O}(\log N)$.

219f0b, 21 lines

```
struct FT {
    vector<ll> s;
    FT(int n) : s(n) {}
    void update(int pos, ll dif) { // a[pos] += dif
        for (; pos < sz(s); pos |= pos + 1) s[pos] += dif;
    }
    ll query(int pos) { // sum of values in [0, pos)
        ll res = 0;
        for (; pos > 0; pos &= pos - 1) res += s[pos-1];
        return res;
    }
    int lower_bound(ll sum) { // min pos st sum of [0, pos] >= sum
        // Returns n if no sum is >= sum, or -1 if empty sum is.
        if (sum <= 0) return -1;
        int pos = 0;
        for (int pw = 1 << 25; pw; pw >>= 1)
            if (pos + pw <= sz(s) && s[pos + pw - 1] < sum)
                pos += pw, sum -= s[pos-1];
        return pos;
    }
};
```

SubMatrix.h

Description: 在给定左上角和右下角的情况下, 快速计算子矩阵和。坐标采用左闭右开区间。

Usage: `SubMatrix<int> m(matrix);`

`m.sum(0, 0, 2, 2);` // 左上角四个元素

Time: $\mathcal{O}(N^2 + Q)$

c59ada, 13 lines

```
template<class T>
struct SubMatrix {
    vector<vector<T>> p;
    SubMatrix(vector<vector<T>>& v) {
        int R = sz(v), C = sz(v[0]);
        p.assign(R+1, vector<T>(C+1));
        rep(r, 0, R) rep(c, 0, C)
            p[r+1][c+1] = v[r][c] + p[r][c+1] + p[r+1][c] - p[r][c];
```

```

}
T sum(int u, int l, int d, int r) {
    return p[d][r] - p[d][l] - p[u][r] + p[u][l];
}
};

```

Numerical (4)

4.1 Polynomials and recurrences

Polynomial.h

c9b7b0, 17 lines

```

struct Poly {
    vector<double> a;
    double operator()(double x) const {
        double val = 0;
        for (int i = sz(a); i--;) (val *= x) += a[i];
        return val;
    }
    void diff() {
        rep(i, 1, sz(a)) a[i-1] = i*a[i];
        a.pop_back();
    }
    void divroot(double x0) {
        double b = a.back(), c; a.back() = 0;
        for(int i=sz(a)-1; i--;) c = a[i], a[i] = a[i+1]*x0+b, b=c;
        a.pop_back();
    }
};

```

PolyRoots.h

Description: Finds the real roots to a polynomial.

Usage: polyRoots({{2,-3,1}},-1e9,1e9) // solve $x^2-3x+2=0$

Time: $\mathcal{O}(n^2 \log(1/\epsilon))$

"Polynomial.h" b00bfe, 23 lines

```

vector<double> polyRoots(Poly p, double xmin, double xmax) {
    if (sz(p.a) == 2) { return {-p.a[0]/p.a[1]}; }
    vector<double> ret;
    Poly der = p;
    der.diff();
    auto dr = polyRoots(der, xmin, xmax);
    dr.push_back(xmin-1);
    dr.push_back(xmax+1);
    sort(all(dr));
    rep(i, 0, sz(dr)-1) {
        double l = dr[i], h = dr[i+1];
        bool sign = p(l) > 0;
        if (sign ^ (p(h) > 0)) {
            rep(it, 0, 60) { // while (h - l > 1e-8)
                double m = (l + h) / 2, f = p(m);
                if ((f <= 0) ^ sign) l = m;
                else h = m;
            }
            ret.push_back((l + h) / 2);
        }
    }
    return ret;
}

```

PolyInterpolate.h

Description: Given n points $(x[i], y[i])$, computes an $n-1$ -degree polynomial p that passes through them: $p(x) = a[0] * x^0 + \dots + a[n-1] * x^{n-1}$. For numerical precision, pick $x[k] = c * \cos(k/(n-1) * \pi)$, $k = 0 \dots n-1$.

Time: $\mathcal{O}(n^2)$

08bf48, 13 lines

```

typedef vector<double> vd;
vd interpolate(vd x, vd y, int n) {
    vd res(n), temp(n);

```

```

rep(k, 0, n-1) rep(i, k+1, n)
    y[i] = (y[i] - y[k]) / (x[i] - x[k]);
double last = 0; temp[0] = 1;
rep(k, 0, n) rep(i, 0, n) {
    res[i] += y[k] * temp[i];
    swap(last, temp[i]);
    temp[i] -= last * x[k];
}
return res;
}

```

BerlekampMassey.h

Description: Recovers any n -order linear recurrence relation from the first $2n$ terms of the recurrence. Useful for guessing linear recurrences after brute-forcing the first terms. Should work on any field, but numerical stability for floats is not guaranteed. Output will have size $\leq n$.

Usage: berlekampMassey({0, 1, 1, 3, 5, 11}) // {1, 2}

Time: $\mathcal{O}(N^2)$

"..number-theory/ModPow.h"

96548b, 20 lines

```

vector<ll> berlekampMassey(vector<ll> s) {
    int n = sz(s), L = 0, m = 0;
    vector<ll> C(n), B(n), T;
    C[0] = B[0] = 1;

    ll b = 1;
    rep(i, 0, n) { ++m;
        ll d = s[i] % mod;
        rep(j, 1, L+1) d = (d + C[j] * s[i - j]) % mod;
        if (!d) continue;
        T = C; ll coef = d * modpow(b, mod-2) % mod;
        rep(j, m, n) C[j] = (C[j] - coef * B[j - m]) % mod;
        if (2 * L > i) continue;
        L = i + 1 - L; B = T; b = d; m = 0;
    }

    C.resize(L + 1); C.erase(C.begin());
    for (ll& x : C) x = (mod - x) % mod;
    return C;
}

```

LinearRecurrence.h

Description: Generates the k 'th term of an n -order linear recurrence $S[i] = \sum_j S[i-j-1]tr[j]$, given $S[0 \dots \geq n-1]$ and $tr[0 \dots n-1]$. Faster than matrix multiplication. Useful together with Berlekamp-Massey.

Usage: linearRec({0, 1}, {1, 1}, k) // k'th Fibonacci number

Time: $\mathcal{O}(n^2 \log k)$

f4e444, 26 lines

```

typedef vector<ll> Poly;
ll linearRec(Poly S, Poly tr, ll k) {
    int n = sz(tr);

    auto combine = [&](Poly a, Poly b) {
        Poly res(n * 2 + 1);
        rep(i, 0, n+1) rep(j, 0, n+1)
            res[i + j] = (res[i + j] + a[i] * b[j]) % mod;
        for (int i = 2 * n; i > n; --i) rep(j, 0, n)
            res[i - 1 - j] = (res[i - 1 - j] + res[i] * tr[j]) % mod;
        res.resize(n + 1);
        return res;
    };

    Poly pol(n + 1), e(pol);
    pol[0] = e[1] = 1;

    for (++k; k; k != 2) {
        if (k % 2) pol = combine(pol, e);
        e = combine(e, e);
    }
}

```

```

ll res = 0;
rep(i, 0, n) res = (res + pol[i + 1] * S[i]) % mod;
return res;
}

```

4.2 Optimization

GoldenSectionSearch.h

Description: Finds the argument minimizing the function f in the interval $[a, b]$ assuming f is unimodal on the interval, i.e. has only one local minimum and no local maximum. The maximum error in the result is eps . Works equally well for maximization with a small change in the code. See TernarySearch.h in the Various chapter for a discrete version.

Usage: double func(double x) { return 4+x+.3*x*x; }

double xmin = gss(-1000, 1000, func);

Time: $\mathcal{O}(\log((b-a)/\epsilon))$

31d45b, 14 lines

```

double gss(double a, double b, double (*f)(double)) {
    double r = (sqrt(5)-1)/2, eps = 1e-7;
    double x1 = b - r*(b-a), x2 = a + r*(b-a);
    double f1 = f(x1), f2 = f(x2);
    while (b-a > eps)
        if (f1 < f2) { //change to > to find maximum
            b = x2; x2 = x1; f2 = f1;
            x1 = b - r*(b-a); f1 = f(x1);
        } else {
            a = x1; x1 = x2; f1 = f2;
            x2 = a + r*(b-a); f2 = f(x2);
        }
    return a;
}

```

HillClimbing.h

Description: Poor man's optimization for unimodal functions

Seecef, 14 lines

```
typedef array<double, 2> P;
```

```

template<class F> pair<double, P> hillClimb(P start, F f) {
    pair<double, P> cur(f(start), start);
    for (double jmp = 1e9; jmp > 1e-20; jmp /= 2) {
        rep(j, 0, 100) rep(dx, -1, 2) rep(dy, -1, 2) {
            P p = cur.second;
            p[0] += dx*jmp;
            p[1] += dy*jmp;
            cur = min(cur, make_pair(f(p), p));
        }
    }
    return cur;
}

```

Integrate.h

Description: Simple integration of a function over an interval using Simpson's rule. The error should be proportional to h^4 , although in practice you will want to verify that the result is stable to desired precision when epsilon changes.

4756fc, 7 lines

```

template<class F>
double quad(double a, double b, F f, const int n = 1000) {
    double h = (b - a) / 2 / n, v = f(a) + f(b);
    rep(i, 1, n*2)
        v += f(a + i*h) * (i&1 ? 4 : 2);
    return v * h / 3;
}

```

IntegrateAdaptive.h

Description: Fast integration using an adaptive Simpson's rule.


```
Usage: double sphereVolume = quad(-1, 1, [](double x) {
return quad(-1, 1, [&](double y) {
return quad(-1, 1, [&](double z) {
return x*x + y*y + z*z < 1; });});});
92dd79, 15 lines
```

```
typedef double d;
#define S(a,b) (f(a) + 4*f((a+b) / 2) + f(b)) * (b-a) / 6

template <class F>
d rec(F& f, d a, d b, d eps, d S) {
    d c = (a + b) / 2;
    d S1 = S(a, c), S2 = S(c, b), T = S1 + S2;
    if (abs(T - S) <= 15 * eps || b - a < 1e-10)
        return T + (T - S) / 15;
    return rec(f, a, c, eps / 2, S1) + rec(f, c, b, eps / 2, S2);
}

template<class F>
d quad(d a, d b, F f, d eps = 1e-8) {
    return rec(f, a, b, eps, S(a, b));
}
```

Simplex.h

Description: Solves a general linear maximization problem: maximize $c^T x$ subject to $Ax \leq b$, $x \geq 0$. Returns $-\text{inf}$ if there is no solution, inf if there are arbitrarily good solutions, or the maximum value of $c^T x$ otherwise. The input vector is set to an optimal x (or in the unbounded case, an arbitrary solution fulfilling the constraints). Numerical stability is not guaranteed. For better performance, define variables such that $x = 0$ is viable.

Usage: `vvd A = {{1,-1}, {-1,1}, {-1,-2}};`
`vvd b = {1,1,-4}, c = {-1,-1}, x;`
`T val = LPSolver(A, b, c).solve(x);`
Time: $\mathcal{O}(NM * \# \text{pivots})$, where a pivot may be e.g. an edge relaxation. $\mathcal{O}(2^n)$ in the general case.

```
aa8530, 68 lines
typedef double T; // long double, Rational, double + mod<P>...
typedef vector<T> vd;
typedef vector<vd> vvd;
```

```
const T eps = 1e-8, inf = 1/.0;
#define MP make_pair
#define ltj(X) if(s == -1 || MP(X[j],N[j]) < MP(X[s],N[s])) s=j
```

```
struct LPSolver {
    int m, n;
    vi N, B;
    vvd D;

    LPSolver(const vvd& A, const vd& b, const vd& c) :
        m(sz(b)), n(sz(c)), N(n+1), B(m), D(m+2, vd(n+2)) {
            rep(i,0,m) rep(j,0,n) D[i][j] = A[i][j];
            rep(i,0,m) { B[i] = n+i; D[i][n] = -1; D[i][n+1] = b[i]; }
            rep(j,0,n) { N[j] = j; D[m][j] = -c[j]; }
            N[n] = -1; D[m+1][n] = 1;
        }
}
```

```
void pivot(int r, int s) {
    T *a = D[r].data(), inv = 1 / a[s];
    rep(i,0,m+2) if (i != r && abs(D[i][s]) > eps) {
        T *b = D[i].data(), inv2 = b[s] * inv;
        rep(j,0,n+2) b[j] -= a[j] * inv2;
        b[s] = a[s] * inv2;
    }
    rep(j,0,n+2) if (j != s) D[r][j] *= inv;
    rep(i,0,m+2) if (i != r) D[i][s] *= -inv;
    D[r][s] = inv;
    swap(B[r], N[s]);
}
```

```
bool simplex(int phase) {
```

Simplex euclid CRT ModularArithmetic

```
int x = m + phase - 1;
for (;;) {
    int s = -1;
    rep(j,0,n+1) if (N[j] != -phase) ltj(D[x]);
    if (D[x][s] >= -eps) return true;
    int r = -1;
    rep(i,0,m) {
        if (D[i][s] <= eps) continue;
        if (r == -1 || MP(D[i][n+1] / D[i][s], B[i])
            < MP(D[r][n+1] / D[r][s], B[r])) r = i;
    }
    if (r == -1) return false;
    pivot(r, s);
}

T solve(vd &x) {
    int r = 0;
    rep(i,1,m) if (D[i][n+1] < D[r][n+1]) r = i;
    if (D[r][n+1] < -eps) {
        pivot(r, n);
        if (!simplex(2) || D[m+1][n+1] < -eps) return -inf;
        rep(i,0,m) if (B[i] == -1) {
            int s = 0;
            rep(j,1,n+1) ltj(D[i]);
            pivot(i, s);
        }
    }
    bool ok = simplex(1); x = vd(n);
    rep(i,0,m) if (B[i] < n) x[B[i]] = D[i][n+1];
    return ok ? D[m][n+1] : inf;
}
};
```

Number theory (5)

5.1 整除性

欧几里得

$$\gcd(a, b) = 1$$

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

找符合 $ax + by = d = \gcd(a, b)$ 的一组 x, y

$$a \cdot a_{\text{inv}} + py = 1 = \gcd(a, p)$$

$$by + (a \bmod b)x = d$$

$$ax + b(y - \left\lfloor \frac{a}{b} \right\rfloor x) = d$$

euclid.h

Description: 找到两个整数 x 和 y , 使得 $ax + by = \gcd(a, b)$ 。如果你只需要 $\gcd$, 可以使用内置的 `__gcd`。如果 a 和 b 互质, 那么 x 就是 $a \bmod b$ 的逆元。
Time: $\mathcal{O}(\log \min(a, b))$

```
33ba8f, 5 lines
ll euclid(ll a, ll b, ll &x, ll &y) {
    if (!b) return x = 1, y = 0, a;
    ll d = euclid(b, a % b, y, x);
    return y -= a/b * x, d;
}
```

中国剩余定理

定理 CRT. 设 $m_1, m_2, \dots, m_n$ 两两互质, 且

$$(S): \begin{cases} x \equiv a_1 \pmod{m_1}, \\ x \equiv a_2 \pmod{m_2}, \\ \vdots \\ x \equiv a_n \pmod{m_n}. \end{cases}$$

令

$$M = \prod_{i=1}^n m_i, \quad M_i = \frac{M}{m_i}, \quad t_i \equiv M_i^{-1} \pmod{m_i}.$$

则方程组 (S) 有解, 并且一组解为

$$x \equiv \sum_{i=1}^n a_i t_i M_i \pmod{M}.$$

构造性证明. 由于 $m_1, m_2, \dots, m_n$ 两两互质, 对任意 i , 有

$$\gcd(M_i, m_i) = 1.$$

因此存在 t_i , 使得

$$t_i M_i \equiv 1 \pmod{m_i}.$$

令

$$x = \sum_{k=1}^n a_k t_k M_k.$$

固定任意 i 。当 $k \neq i$ 时, 由于 $m_i \mid M_k$, 有

$$a_k t_k M_k \equiv 0 \pmod{m_i}.$$

于是

$$\begin{aligned} x &= a_i t_i M_i + \sum_{k \neq i} a_k t_k M_k \\ &\equiv a_i \cdot 1 + \sum_{k \neq i} 0 \pmod{m_i} \\ &\equiv a_i \pmod{m_i}. \end{aligned}$$

因此 x 同时满足所有同余式, 故 x 是方程组 (S) 的一组解。

CRT.h

Description: 中国剩余定理。
`crt(a, m, b, n)` 计算 x , 使得 $x \equiv a \pmod{m}$, $x \equiv b \pmod{n}$ 。如果 $|a| < m$ 且 $|b| < n$, 则 x 将满足 $0 \leq x < \text{lcm}(m, n)$ 。假设 $mn < 2^{62}$ 。

Time: $\log(n)$

```
04d93a, 7 lines
"euclid.h"
ll crt(ll a, ll m, ll b, ll n) {
    if (n > m) swap(a, b), swap(m, n);
    ll x, y, g = euclid(m, n, x, y);
    assert((a - b) % g == 0); // else no solution
    x = (b - a) % n * x % n / g * m + a;
    return x < 0 ? x + m*n/g : x;
}
```

5.2 模运算

ModularArithmetic.h

Description: 用于模运算, 设置 MOD 后使用。

```
2f9ed0, 18 lines
"euclid.h"
constexpr ll MOD = 17; // change to something else
struct Mod {
    ll x;
    Mod(ll _x) : x((_x % MOD + MOD) % MOD) {}
}
```

```

Mod operator+(Mod b) { return Mod(((x + b.x)%MOD+MOD)%MOD); }
Mod operator-(Mod b) { return Mod(((x - b.x)%MOD+MOD)%MOD); }
Mod operator*(Mod b) { return Mod((x * b.x %MOD+MOD)%MOD); }
Mod operator/(Mod b) { return *this * invert(b); }
Mod invert(Mod a) {
    ll x, y, g = euclid(a.x, MOD, x, y);
    assert(g == 1);return Mod((x % MOD + MOD) % MOD);
}
Mod operator^(ll e) {
    if (!e) return Mod(1);
    Mod r = *this ^ (e / 2); r = r * r;
    return e & 1 ? *this * r : r;
}
};

```

模逆元

$$\frac{a}{b} \equiv a \cdot b_{\text{inv}} \equiv x \pmod{p}$$

$$\Rightarrow a \equiv bx \pmod{p}$$

$$\Rightarrow a \cdot b_{\text{inv}} \cdot b \equiv bx \pmod{p}$$

$$\Rightarrow a \cdot (b_{\text{inv}} \cdot b - 1) \equiv 0 \pmod{p}$$

$$\Rightarrow b_{\text{inv}} \cdot b \equiv 1 \pmod{p}$$

$$\text{inv存在} \Leftrightarrow \gcd(b, p) = 1$$

$$b \cdot b_{\text{inv}} = k \cdot p + 1, \quad g = \gcd(b, p) \Rightarrow g \cdot b_1 \cdot b_{\text{inv}} = k \cdot g \cdot p_1 + 1$$

$$\Rightarrow g \cdot (b_1 \cdot b_{\text{inv}} - k \cdot p_1) = 1$$

$$\text{数域为}\mathbb{Z} \Rightarrow g = 1, \quad b_1 \cdot b_{\text{inv}} - k \cdot p_1 = 1$$

ModInverse.h

Description: 预计算模逆元，假设 $\text{LIM} \leq \text{mod}$ 且 mod 是质数。 17b468, 4 lines

```

constexpr ll MOD = 1000000007, LIM = 200000;
vector<int> inv(LIM + 1, 0); inv[1] = 1;
for (int i = 2; i <= LIM; i++)
    inv[i] = MOD - (MOD / i) * inv[MOD % i] % MOD;

```

ModPow.h

const ll mod = 1000000007; // faster if const

```

ll modpow(ll b, ll e) {
    ll ans = 1;
    for (; e; b = b * b % mod, e /= 2)
        if (e & 1) ans = ans * b % mod;
    return ans;
}

```

ModLog.h

Description: Returns the smallest $x > 0$ s.t. $a^x = b \pmod{m}$, or -1 if no such x exists. $\text{modLog}(a,1,m)$ can be used to calculate the order of a .

Time: $\mathcal{O}(\sqrt{m})$ c040b8, 11 lines

```

ll modLog(ll a, ll b, ll m) {
    ll n = (ll) sqrt(m) + 1, e = 1, f = 1, j = 1;
    unordered_map<ll, ll> A;
    while (j <= n && (e = f = e * a % m) != b % m)
        A[e * b % m] = j++;
    if (e == b % m) return j;
    if (__gcd(m, e) == __gcd(m, b))
        rep(i, 2, n+2) if (A.count(e = e * f % m))
            return n * i - A[e];
    return -1;
}

```

ModInverse ModPow ModLog IntPerm multinomial

Combinatorial (6)

6.1 Permutations

6.1.1 Factorial

n	1	2	3	4	5	6	7	8	9	10
$n!$	1	2	6	24	120	720	5040	40320	362880	3628800
n	11	12	13	14	15	16	17			
$n!$	4.0e7	4.8e8	6.2e9	8.7e10	1.3e12	2.1e13	3.6e14			
n	20	25	30	40	50	100	150	171		
$n!$	2e18	2e25	3e32	8e47	3e64	9e157	6e262	>DBL_MAX		

IntPerm.h

Description: Permutation -> integer conversion. (Not order preserving.) Integer -> permutation can use a lookup table.

Time: $\mathcal{O}(n)$ 044568, 6 lines

```

int IntPerm(Cycle& c, v) {
    int use = 0, i = 0, r = 0;
    for (int j = 0; j < c.size(); j++) {
        int p = c[j];
        if (p < use) continue;
        use = p;
        r = (r * p) % MOD;
    }
    return r;
}

```

6.1.2 Cycles n -permutations whose cycle lengths are $1, 2, \dots, n$. $n!$ Derangements of n -permutations whose cycle lengths are $2, 3, \dots, n$. $n!$ Burnside's lemma Given a group G of symmetries and a set X , the number of elements of X up to symmetry equals $\frac{1}{|G|} \sum_{g \in G} \text{fix}(g)$

where X^g are the elements fixed by g ($g \cdot x = x$).

If $f(n)$ counts “configurations” (of some sort) of length n , we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n} \sum_{k=0}^{n-1} f(\gcd(n, k)) = \frac{1}{n} \sum_{k|n} f(k) \phi(n/k).$$

6.2 Partitions and subsets

6.2.1 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1, \quad p(n) = \sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n - k(3k - 1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100	
$p(n)$	1	1	1	2	3	5	7	11	15	22	30	627	$\sim 2e5$	$\sim 2e8$

6.2.2 Lucas' Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + \dots + n_1 p + n_0$ and $m = m_k p^k + \dots + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^k \binom{n_i}{m_i} \pmod{p}$.

6.2.3 Binomials

multinomial.h

Description: Computes $\binom{k_1 + \dots + k_n}{k_1, k_2, \dots, k_n} = \frac{(\sum k_i)!}{k_1! k_2! \dots k_n!}$. a0a312, 5 lines

```

ll multinomial(vi& v) {
    ll c = 1, m = v.empty() ? 1 : v[0];
    rep(i, 1, sz(v)) rep(j, 0, v[i]) c = c * ++m / (j+1);
    return c;
}

```

6.3 General purpose numbers

6.3.1 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t - 1}$ (FFT-able).

$$B[0, \dots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \dots]$$

Sums of powers:

$$\sum_{i=1}^n n^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k \cdot (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\sum_{i=m}^{\infty} f(i) = \int_m^{\infty} f(x) dx - \sum_{k=1}^{\infty} \frac{B_k}{k!} f^{(k-1)}(m)$$

$$\approx \int_m^{\infty} f(x) dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m))$$

6.3.2 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k), \quad c(0, 0) = 1$$

$$\sum_{k=0}^n c(n, k) x^k = x(x+1) \dots (x+n-1)$$

$$c(8, k) = 8, 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$$

$$c(n, 2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \dots$$

6.3.3 Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k j:s s.t. $\pi(j) > \pi(j+1)$, $k+1$ j:s s.t. $\pi(j) \geq j$, k j:s s.t. $\pi(j) > j$.

$$E(n, k) = (n-k)E(n-1, k-1) + (k+1)E(n-1, k)$$

$$E(n, 0) = E(n, n-1) = 1$$

$$E(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

6.3.4 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n, k) = S(n-1, k-1) + kS(n-1, k)$$

$$S(n, 1) = S(n, n) = 1$$

$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

6.3.5 Bell numbers

Total number of partitions of n distinct elements. $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \dots$. For p prime,

$$B(p^m + n) \equiv mB(n) + B(n+1) \pmod{p}$$

6.3.6 Labeled unrooted trees

on n vertices: n^{n-2}

on k existing trees of size n_i : $n_1 n_2 \dots n_k n^{k-2}$

with degrees d_i : $(n-2)! / ((d_1-1)! \dots (d_n-1)!)$

6.3.7 Catalan numbers

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1} = \frac{(2n)!}{(n+1)n!}$$

$$C_0 = 1, C_{n+1} = \frac{2(2n+1)}{n+2} C_n, C_{n+1} = \sum C_i C_{n-i}$$

$$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \dots$$

- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n+1$ leaves (0 or 2 children).
- ordered trees with $n+1$ vertices.
- ways a convex polygon with $n+2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

Graph (7)

7.1 Fundamentals

BellmanFord.h

Description: Calculates shortest paths from s in a graph that might have negative edge weights. Unreachable nodes get $\text{dist} = \text{inf}$; nodes reachable through negative-weight cycles get $\text{dist} = -\text{inf}$. Assumes $V^2 \max |w_i| < \sim 2^{63}$.
Time: $\mathcal{O}(VE)$

830a8f, 23 lines

```
const ll inf = LLONG_MAX;
struct Ed { int a, b, w, s() { return a < b ? a : -a; } };
struct Node { ll dist = inf; int prev = -1; };
```

```
void bellmanFord(vector<Node>& nodes, vector<Ed>& eds, int s) {
    nodes[s].dist = 0;
    sort(all(eds), [](Ed a, Ed b) { return a.s() < b.s(); });
```

```
    int lim = sz(nodes) / 2 + 2; // /3+100 with shuffled vertices
    rep(i, 0, lim) for (Ed ed : eds) {
        Node cur = nodes[ed.a], &dest = nodes[ed.b];
        if (abs(cur.dist) == inf) continue;
        ll d = cur.dist + ed.w;
```

```
        if (d < dest.dist) {
            dest.prev = ed.a;
            dest.dist = (i < lim-1 ? d : -inf);
        }
    }
    rep(i, 0, lim) for (Ed e : eds) {
        if (nodes[e.a].dist == -inf)
            nodes[e.b].dist = -inf;
    }
}
```

FloydWarshall.h

Description: Calculates all-pairs shortest path in a directed graph that might have negative edge weights. Input is an distance matrix m , where $m[i][j] = \text{inf}$ if i and j are not adjacent. As output, $m[i][j]$ is set to the shortest distance between i and j , inf if no path, or $-\text{inf}$ if the path goes through a negative-weight cycle.

Time: $\mathcal{O}(N^3)$

531245, 12 lines

```
const ll inf = 1LL << 62;
void floydWarshall(vector<vector<ll>>& m) {
    int n = sz(m);
    rep(i, 0, n) m[i][i] = min(m[i][i], 0LL);
    rep(k, 0, n) rep(i, 0, n) rep(j, 0, n)
        if (m[i][k] != inf && m[k][j] != inf) {
            auto newDist = max(m[i][k] + m[k][j], -inf);
            m[i][j] = min(m[i][j], newDist);
        }
    rep(k, 0, n) if (m[k][k] < 0) rep(i, 0, n) rep(j, 0, n)
        if (m[i][k] != inf && m[k][j] != inf) m[i][j] = -inf;
}
```

TopoSort.h

Description: Topological sorting. Given is an oriented graph. Output is an ordering of vertices, such that there are edges only from left to right. If there are cycles, the returned list will have size smaller than n – nodes reachable from cycles will not be returned.

Time: $\mathcal{O}(|V| + |E|)$

d678d8, 8 lines

```
vi topoSort(const vector<vi>& gr) {
    vi indeg(sz(gr)), q;
    for (auto& li : gr) for (int x : li) indeg[x]++;
    rep(i, 0, sz(gr)) if (indeg[i] == 0) q.push_back(i);
    rep(j, 0, sz(q)) for (int x : gr[q[j]])
        if (--indeg[x] == 0) q.push_back(x);
    return q;
}
```

7.2 Network flow

PushRelabel.h

Description: Push-relabel using the highest label selection rule and the gap heuristic. Quite fast in practice. To obtain the actual flow, look at positive values only.

Time: $\mathcal{O}(V^2\sqrt{E})$

0ae1d4, 48 lines

```
struct PushRelabel {
    struct Edge {
        int dest, back;
        ll f, c;
    };
    vector<vector<Edge>>> g;
    vector<ll> ec;
    vector<Edge*> cur;
    vector<vi> hs; vi H;
    PushRelabel(int n) : g(n), ec(n), cur(n), hs(2*n), H(n) {}

    void addEdge(int s, int t, ll cap, ll rcap=0) {
        if (s == t) return;
```

```
        g[s].push_back({t, sz(g[t]), 0, cap});
        g[t].push_back({s, sz(g[s])-1, 0, rcap});
    }
```

```
void addFlow(Edge& e, ll f) {
    Edge &back = g[e.dest][e.back];
    if (!ec[e.dest] && f) hs[H[e.dest]].push_back(e.dest);
    e.f += f; e.c -= f; ec[e.dest] += f;
    back.f -= f; back.c += f; ec[back.dest] -= f;
}
```

```
ll calc(int s, int t) {
    int v = sz(g); H[s] = v; ec[t] = 1;
    vi co(2*v); co[0] = v-1;
    rep(i, 0, v) cur[i] = g[i].data();
    for (Edge& e : g[s]) addFlow(e, e.c);
```

```
    for (int hi = 0;;) {
        while (hs[hi].empty()) if (!hi--) return -ec[s];
        int u = hs[hi].back(); hs[hi].pop_back();
        while (ec[u] > 0) // discharge u
            if (cur[u] == g[u].data() + sz(g[u])) {
                H[u] = 1e9;
                for (Edge& e : g[u]) if (e.c && H[u] > H[e.dest]+1)
                    H[u] = H[e.dest]+1, cur[u] = &e;
                if (++co[H[u]], !--co[hi] && hi < v)
                    rep(i, 0, v) if (hi < H[i] && H[i] < v)
                        --co[H[i]], H[i] = v + 1;
                hi = H[u];
            } else if (cur[u]->c && H[u] == H[cur[u]->dest]+1)
                addFlow(*cur[u], min(ec[u], cur[u]->c));
            else ++cur[u];
    }
    bool leftOfMinCut(int a) { return H[a] >= sz(g); }
};
```

MinCostMaxFlow.h

Description: Min-cost max-flow. If costs can be negative, call setpi before maxflow, but note that negative cost cycles are not supported. To obtain the actual flow, look at positive values only.

Time: $\mathcal{O}(FE \log(V))$ where F is max flow. $\mathcal{O}(VE)$ for setpi.

58385b, 79 lines

```
#include <bits/extc++.h>
```

```
const ll INF = numeric_limits<ll>::max() / 4;
```

```
struct MCMF {
    struct edge {
        int from, to, rev;
        ll cap, cost, flow;
    };
    int N;
    vector<vector<edge>>> ed;
    vi seen;
    vector<ll> dist, pi;
    vector<edge*> par;
```

```
MCMF(int N) : N(N), ed(N), seen(N), dist(N), pi(N), par(N) {}
```

```
void addEdge(int from, int to, ll cap, ll cost) {
    if (from == to) return;
    ed[from].push_back(edge{ from, to, sz(ed[to]), cap, cost, 0 });
    ed[to].push_back(edge{ to, from, sz(ed[from])-1, 0, -cost, 0 });
}
```

```
void path(int s) {
    fill(all(seen), 0);
    fill(all(dist), INF);
    dist[s] = 0; ll di;
```



```

__gnu_pbds::priority_queue<pair<ll, int>> q;
vector<decltype(q)::point_iterator> its(N);
q.push({ 0, s });

while (!q.empty()) {
    s = q.top().second; q.pop();
    seen[s] = 1; di = dist[s] + pi[s];
    for (edge& e : ed[s]) if (!seen[e.to]) {
        ll val = di - pi[e.to] + e.cost;
        if (e.cap - e.flow > 0 && val < dist[e.to]) {
            dist[e.to] = val;
            par[e.to] = &e;
            if (its[e.to] == q.end())
                its[e.to] = q.push({ -dist[e.to], e.to });
            else
                q.modify(its[e.to], { -dist[e.to], e.to });
        }
    }
}
rep(i,0,N) pi[i] = min(pi[i] + dist[i], INF);
}

pair<ll, ll> maxflow(int s, int t) {
    ll totflow = 0, totcost = 0;
    while (path(s, seen[t])) {
        ll fl = INF;
        for (edge* x = par[t]; x; x = par[x->from])
            fl = min(fl, x->cap - x->flow);

        totflow += fl;
        for (edge* x = par[t]; x; x = par[x->from]) {
            x->flow += fl;
            ed[x->to][x->rev].flow -= fl;
        }
    }
    rep(i,0,N) for(edge& e : ed[i]) totcost += e.cost * e.flow;
    return {totflow, totcost/2};
}

// If some costs can be negative, call this before maxflow:
void setpi(int s) { // (otherwise, leave this out)
    fill(all(pi), INF); pi[s] = 0;
    int it = N, ch = 1; ll v;
    while (ch-- && it--)
        rep(i,0,N) if (pi[i] != INF)
            for (edge& e : ed[i]) if (e.cap)
                if ((v = pi[i] + e.cost) < pi[e.to])
                    pi[e.to] = v, ch = 1;
    assert(it >= 0); // negative cost cycle
}
};

```

EdmondsKarp.h

Description: Flow algorithm with guaranteed complexity $O(VE^2)$. To get edge flow values, compare capacities before and after, and take the positive values only.

482fe0, 36 lines

```

template<class T> T edmondsKarp(vector<unordered_map<int, T>>&
    graph, int source, int sink) {
    assert(source != sink);
    T flow = 0;
    vi par(sz(graph)), q = par;

    for (;;) {
        fill(all(par), -1);
        par[source] = 0;
        int ptr = 1;
        q[0] = source;

```

```

        rep(i,0,ptr) {
            int x = q[i];
            for (auto e : graph[x]) {
                if (par[e.first] == -1 && e.second > 0) {
                    par[e.first] = x;
                    q[ptr++] = e.first;
                    if (e.first == sink) goto out;
                }
            }
        }
        return flow;
    out:
    T inc = numeric_limits<T>::max();
    for (int y = sink; y != source; y = par[y])
        inc = min(inc, graph[par[y]][y]);

    flow += inc;
    for (int y = sink; y != source; y = par[y]) {
        int p = par[y];
        if ((graph[p][y] -= inc) <= 0) graph[p].erase(y);
        graph[y][p] += inc;
    }
}

```

MinCut.h

Description: After running max-flow, the left side of a min-cut from s to t is given by all vertices reachable from s , only traversing edges with positive residual capacity.

GlobalMinCut.h

Description: Find a global minimum cut in an undirected graph, as represented by an adjacency matrix.

Time: $O(V^3)$

8b0e19, 21 lines

```

pair<int, vi> globalMinCut(vector<vi> mat) {
    pair<int, vi> best = {INT_MAX, {}};
    int n = sz(mat);
    vector<vi> co(n);
    rep(i,0,n) co[i] = {i};
    rep(ph,1,n) {
        vi w = mat[0];
        size_t s = 0, t = 0;
        rep(it,0,n-ph) { //  $O(V^2) \rightarrow O(E \log V)$  with prio. queue
            w[t] = INT_MIN;
            s = t, t = max_element(all(w)) - w.begin();
            rep(i,0,n) w[i] += mat[t][i];
        }
        best = min(best, {w[t] - mat[t][t], co[t]});
        co[s].insert(co[s].end(), all(co[t]));
        rep(i,0,n) mat[s][i] += mat[t][i];
        rep(i,0,n) mat[i][s] = mat[s][i];
        mat[0][t] = INT_MIN;
    }
    return best;
}

```

GomoryHu.h

Description: Given a list of edges representing an undirected flow graph, returns edges of the Gomory-Hu tree. The max flow between any pair of vertices is given by minimum edge weight along the Gomory-Hu tree path.

Time: $O(V)$ Flow Computations

"PushRelabel.h"

0418b3, 13 lines

```

typedef array<ll, 3> Edge;
vector<Edge> gomoryHu(int N, vector<Edge> ed) {
    vector<Edge> tree;

```

```

    vi par(N);
    rep(i,1,N) {
        PushRelabel D(N); // Dinic also works
        for (Edge t : ed) D.addEdge(t[0], t[1], t[2], t[2]);
        tree.push_back({i, par[i], D.calc(i, par[i])});
        rep(j,i+1,N)
            if (par[j] == par[i] && D.leftOfMinCut(j)) par[j] = i;
    }
    return tree;
}

```

7.3 Matching

HopcroftKarp.h

Description: Fast bipartite matching algorithm. Graph g should be a list of neighbors of the left partition, and r should be a vector full of -1 's of the same size as the right partition. Returns the size of the matching. $r[i]$ will be the match for vertex i on the right side, or -1 if it's not matched.

Time: $O(E\sqrt{V})$

731cfb, 20 lines

```

int hopcroftKarp(vector<vi>& g, vi& r) {
    int n = sz(g), res = 0;
    vi l(n, -1), q(n), d(n);
    auto dfs = [&](auto f, int u) -> bool {
        int t = exchange(d[u], 0) + 1;
        for (int v : g[u])
            if (r[v] == -1 || (d[r[v]] == t && f(f, r[v])))
                return l[u] = v, r[v] = u, 1;
        return 0;
    };
    for (int t = 0, f = 0; t = f = 0, d.assign(n, 0)) {
        rep(i,0,n) if (l[i] == -1) q[t++] = i, d[i] = 1;
        rep(i,0,t) for (int v : g[q[i]]) {
            if (r[v] == -1) f = 1;
            else if (d[r[v]] > d[q[i]] + 1, q[t++] = r[v];
        }
        if (!f) return res;
        rep(i,0,n) if (l[i] == -1) res += dfs(dfs, i);
    }
}

```

DFSMatching.h

Description: Simple bipartite matching algorithm. Graph g should be a list of neighbors of the left partition, and $btoa$ should be a vector full of -1 's of the same size as the right partition. Returns the size of the matching. $btoa[i]$ will be the match for vertex i on the right side, or -1 if it's not matched.

Usage: `vi btoa(m, -1); dfsMatching(g, btoa);`

Time: $O(VE)$

522b98, 22 lines

```

bool find(int j, vector<vi>& g, vi& btoa, vi& vis) {
    if (btoa[j] == -1) return 1;
    vis[j] = 1; int di = btoa[j];
    for (int e : g[di])
        if (!vis[e] && find(e, g, btoa, vis)) {
            btoa[e] = di;
            return 1;
        }
    return 0;
}

int dfsMatching(vector<vi>& g, vi& btoa) {
    vi vis;
    rep(i,0,sz(g)) {
        vis.assign(sz(btoa), 0);
        for (int j : g[i])
            if (find(j, g, btoa, vis)) {
                btoa[j] = i;
                break;
            }
    }
}

```

```

return sz(btoa) - (int)count(all(btoa), -1);
}

```

MinimumVertexCover.h

Description: Finds a minimum vertex cover in a bipartite graph. The size is the same as the size of a maximum matching, and the complement is a maximum independent set.

```

"DFSMatching.h"
da4196, 20 lines

vi cover(vector<vi>& g, int n, int m) {
    vi match(m, -1);
    int res = dfsMatching(g, match);
    vector<bool> lfound(n, true), seen(m);
    for (int it : match) if (it != -1) lfound[it] = false;
    vi q, cover;
    rep(i,0,n) if (lfound[i]) q.push_back(i);
    while (!q.empty()) {
        int i = q.back(); q.pop_back();
        lfound[i] = 1;
        for (int e : g[i]) if (!seen[e] && match[e] != -1) {
            seen[e] = true;
            q.push_back(match[e]);
        }
    }
    rep(i,0,n) if (!lfound[i]) cover.push_back(i);
    rep(i,0,m) if (seen[i]) cover.push_back(n+i);
    assert(sz(cover) == res);
    return cover;
}

```

WeightedMatching.h

Description: Given a weighted bipartite graph, matches every node on the left with a node on the right such that no nodes are in two matchings and the sum of the edge weights is minimal. Takes cost[N][M], where cost[i][j] = cost for L[i] to be matched with R[j] and returns (min cost, match), where L[i] is matched with R[match[i]]. Negate costs for max cost. Requires $N \leq M$.
Time: $O(N^2M)$

```

1e0fe9, 31 lines

pair<int, vi> hungarian(const vector<vi>& a) {
    if (a.empty()) return {0, {}};
    int n = sz(a) + 1, m = sz(a[0]) + 1;
    vi u(n), v(m), p(m), ans(n - 1);
    rep(i,1,n) {
        p[0] = i;
        int j0 = 0; // add "dummy" worker 0
        vi dist(m, INT_MAX), pre(m, -1);
        vector<bool> done(m + 1);
        do { // dijkstra
            done[j0] = true;
            int i0 = p[j0], j1, delta = INT_MAX;
            rep(j,1,m) if (!done[j]) {
                auto cur = a[i0 - 1][j - 1] - u[i0] - v[j];
                if (cur < dist[j]) dist[j] = cur, pre[j] = j0;
                if (dist[j] < delta) delta = dist[j], j1 = j;
            }
            rep(j,0,m) {
                if (done[j]) u[p[j]] += delta, v[j] -= delta;
                else dist[j] -= delta;
            }
            j0 = j1;
        } while (p[j0]);
        while (j0) { // update alternating path
            int j1 = pre[j0];
            p[j0] = p[j1], j0 = j1;
        }
    }
    rep(j,1,m) if (p[j]) ans[p[j] - 1] = j - 1;
    return {-v[0], ans}; // min cost
}

```

GeneralMatching.h

Description: Matching for general graphs. Fails with probability N/mod .
Time: $O(N^3)$

```

"./numerical/MatrixInverse-mod.h"
cb1912, 40 lines

vector<pii> generalMatching(int N, vector<pii>& ed) {
    vector<vector<ll>> mat(N, vector<ll>(N)), A;
    for (pii pa : ed) {
        int a = pa.first, b = pa.second, r = rand() % mod;
        mat[a][b] = r, mat[b][a] = (mod - r) % mod;
    }

    int r = matInv(A = mat), M = 2*N - r, fi, fj;
    assert(r % 2 == 0);

    if (M != N) do {
        mat.resize(M, vector<ll>(M));
        rep(i,0,M) {
            mat[i].resize(M);
            rep(j,N,M) {
                int r = rand() % mod;
                mat[i][j] = r, mat[j][i] = (mod - r) % mod;
            }
        } while (matInv(A = mat) != M);

    vi has(M, 1); vector<pii> ret;
    rep(it,0,M/2) {
        rep(i,0,M) if (has[i])
            rep(j,i+1,M) if (A[i][j] && mat[i][j]) {
                fi = i; fj = j; goto done;
            } assert(0); done:
        if (fj < N) ret.emplace_back(fi, fj);
        has[fi] = has[fj] = 0;
        rep(sw,0,2) {
            ll a = modpow(A[fi][fj], mod-2);
            rep(i,0,M) if (has[i] && A[i][fj]) {
                ll b = A[i][fj] * a % mod;
                rep(j,0,M) A[i][j] = (A[i][j] - A[fi][j] * b) % mod;
            }
            swap(fi, fj);
        }
    }
    return ret;
}

```

7.4 DFS algorithms

SCC.h

Description: Finds strongly connected components in a directed graph. If vertices u, v belong to the same component, we can reach u from v and vice versa.

Usage: scc(graph, [&](vi& v) { ... }) visits all components in reverse topological order. comp[i] holds the component index of a node (a component only has edges to components with lower index). ncomps will contain the number of components.

```

Time: O(E + V)
76b5c9, 24 lines

vi val, comp, z, cont;
int Time, ncomps;
template<class G, class F> int dfs(int j, G& g, F& f) {
    int low = val[j] = ++Time, x; z.push_back(j);
    for (auto e : g[j]) if (comp[e] < 0)
        low = min(low, val[e] ? dfs(e, g, f));

    if (low == val[j]) {
        do {
            x = z.back(); z.pop_back();
            comp[x] = ncomps;
            cont.push_back(x);
        } while (x != j);
    }
}

```

```

f(cont); cont.clear();
ncomps++;
}
return val[j] = low;
}
template<class G, class F> void scc(G& g, F f) {
    int n = sz(g);
    val.assign(n, 0); comp.assign(n, -1);
    Time = ncomps = 0;
    rep(i,0,n) if (comp[i] < 0) dfs(i, g, f);
}

```

BiconnectedComponents.h

Description: Finds all biconnected components in an undirected graph, and runs a callback for the edges in each. In a biconnected component there are at least two internally disjoint paths between any two nodes (a cycle exists through them). Note that a node can be in several components. An edge which is not in a component is a bridge, i.e., not part of any cycle.

Usage: int eid = 0; ed.resize(N);
for each edge (a,b) {
ed[a].emplace_back(b, eid);
ed[b].emplace_back(a, eid++); }
bicomps[&](const vi& edgelist) { ... };
Time: $O(E + V)$

```

c6b7c7, 32 lines

vi num, st;
vector<vector<pii>> ed;
int Time;
template<class F>
int dfs(int at, int par, F& f) {
    int me = num[at] = ++Time, top = me;
    for (auto [y, e] : ed[at]) if (e != par) {
        if (num[y]) {
            top = min(top, num[y]);
            if (num[y] < me)
                st.push_back(e);
        } else {
            int si = sz(st);
            int up = dfs(y, e, f);
            top = min(top, up);
            if (up == me) {
                st.push_back(e);
                f(vi(st.begin() + si, st.end()));
                st.resize(si);
            }
        }
        else if (up < me) st.push_back(e);
        else { /* e is a bridge */ }
    }
    return top;
}

```

```

template<class F>
void bicomps(F f) {
    num.assign(sz(ed), 0);
    rep(i,0,sz(ed)) if (!num[i]) dfs(i, -1, f);
}

```

2sat.h

Description: Calculates a valid assignment to boolean variables $a, b, c, \dots$ to a 2-SAT problem, so that an expression of the type $(a||b)\&\&(!a||c)\&\&(d||!b)\&\&\dots$ becomes true, or reports that it is unsatisfiable. Negated variables are represented by bit-inversions ($\sim x$).

Usage: TwoSat ts(number of boolean variables);
ts.either(0, ~3); // Var 0 is true or var 3 is false
ts.setValue(2); // Var 2 is true
ts.atMostOne({0, ~1, 2}); // ≤ 1 of vars 0, ~1 and 2 are true
ts.solve(); // Returns true iff it is solvable
ts.values[0..N-1] holds the assigned values to the vars

Time: $\mathcal{O}(N + E)$, where N is the number of boolean variables, and E is the number of clauses.

5f9706, 56 lines

```
struct TwoSat {
    int N;
    vector<vi> gr;
    vi values; // 0 = false, 1 = true

    TwoSat(int n = 0) : N(n), gr(2*n) {}

    int addVar() { // (optional)
        gr.emplace_back();
        gr.emplace_back();
        return N++;
    }
}
```

```
void either(int f, int j) {
    f = max(2*f, -1-2*f);
    j = max(2*j, -1-2*j);
    gr[f].push_back(j+1);
    gr[j].push_back(f+1);
}

void setValue(int x) { either(x, x); }
```

```
void atMostOne(const vi& li) { // (optional)
    if (sz(li) <= 1) return;
    int cur = ~li[0];
    rep(i, 2, sz(li)) {
        int next = addVar();
        either(cur, ~li[i]);
        either(cur, next);
        either(~li[i], next);
        cur = ~next;
    }
    either(cur, ~li[1]);
}
```

```
vi val, comp, z; int time = 0;
int dfs(int i) {
    int low = val[i] = ++time, x; z.push_back(i);
    for(int e : gr[i]) if (!comp[e])
        low = min(low, val[e] ? dfs(e));
    if (low == val[i]) do {
        x = z.back(); z.pop_back();
        comp[x] = low;
        if (values[x>>1] == -1)
            values[x>>1] = x&1;
    } while (x != i);
    return val[i] = low;
}
```

```
bool solve() {
    values.assign(N, -1);
    val.assign(2*N, 0); comp = val;
    rep(i, 0, 2*N) if (!comp[i]) dfs(i);
    rep(i, 0, N) if (comp[2*i] == comp[2*i+1]) return 0;
    return 1;
}

};
```

EulerWalk.h

Description: Eulerian undirected/directed path/cycle algorithm. Input should be a vector of (dest, global edge index), where for undirected graphs, forward/backward edges have the same index. Returns a list of nodes in the Eulerian path/cycle with src at both start and end, or empty list if no cycle/path exists. To get edge indices back, add .second to s and ret.

Time: $\mathcal{O}(V + E)$

780b64, 15 lines

```
vi eulerWalk(vector<vector<pii>>& gr, int nedges, int src=0) {
    int n = sz(gr);
```

```
vi D(n), its(n), eu(nedges), ret, s = {src};
D[src]++; // to allow Euler paths, not just cycles
while (!s.empty()) {
    int x = s.back(), y, e, &it = its[x], end = sz(gr[x]);
    if (it == end) { ret.push_back(x); s.pop_back(); continue; }
    tie(y, e) = gr[x][it++];
    if (!eu[e]) {
        D[x]--, D[y]++;
        eu[e] = 1; s.push_back(y);
    }
}

for (int x : D) if (x < 0 || sz(ret) != nedges+1) return {};
return {ret.rbegin(), ret.rend()};
}
```

7.5 Coloring

EdgeColoring.h

Description: Given a simple, undirected graph with max degree D , computes a $(D+1)$ -coloring of the edges such that no neighboring edges share a color. (D -coloring is NP-hard, but can be done for bipartite graphs by repeated matchings of max-degree nodes.)

Time: $\mathcal{O}(NM)$

e210e2, 31 lines

```
vi edgeColoring(int N, vector<pii> eds) {
    vi cc(N+1), ret(sz(eds)), fan(N), free(N), loc;
    for (pii e : eds) ++cc[e.first], ++cc[e.second];
    int u, v, ncols = *max_element(all(cc)) + 1;
    vector<vi> adj(N, vi(ncols, -1));
    for (pii e : eds) {
        tie(u, v) = e;
        fan[0] = v;
        loc.assign(ncols, 0);
        int at = u, end = u, d, c = free[u], ind = 0, i = 0;
        while (d = free[v], !loc[d] && (v = adj[u][d]) != -1)
            loc[d] = ++ind, cc[ind] = d, fan[ind] = v;
        cc[loc[d]] = c;
        for (int cd = d; at != -1; cd ^= c ^ d, at = adj[at][cd])
            swap(adj[at][cd], adj[end = at][cd ^ c ^ d]);
        while (adj[fan[i]][d] != -1) {
            int left = fan[i], right = fan[++i], e = cc[i];
            adj[u][e] = left;
            adj[left][e] = u;
            adj[right][e] = -1;
            free[right] = e;
        }
        adj[u][d] = fan[i];
        adj[fan[i]][d] = u;
        for (int y : {fan[0], u, end})
            for (int& z = free[y] = 0; adj[y][z] != -1; z++);
    }
    rep(i, 0, sz(eds))
        for (tie(u, v) = eds[i]; adj[u][ret[i]] != v; ++ret[i];
    return ret;
}
```

7.6 Heuristics

MaximalCliques.h

Description: Runs a callback for all maximal cliques in a graph (given as a symmetric bitset matrix; self-edges not allowed). Callback is given a bitset representing the maximal clique.

Time: $\mathcal{O}(3^{n/3})$, much faster for sparse graphs

b0d5b1, 12 lines

```
typedef bitset<128> B;
template<class F>
void cliques(vector<B>& eds, F f, B P = ~B(), B X={}, B R={}) {
    if (!P.any()) { if (!X.any()) f(R); return; }
    auto q = (P | X)._Find_first();
    auto cands = P & ~eds[q];
```

```
rep(i, 0, sz(eds)) if (cands[i]) {
    R[i] = 1;
    cliques(eds, f, P & eds[i], X & eds[i], R);
    R[i] = P[i] = 0; X[i] = 1;
}
}
```

MaximumClique.h

Description: Quickly finds a maximum clique of a graph (given as symmetric bitset matrix; self-edges not allowed). Can be used to find a maximum independent set by finding a clique of the complement graph.

Time: Runs in about 1s for $n=155$ and worst case random graphs ($p=.90$). Runs faster for sparse graphs.

f7c0bc, 49 lines

```
typedef vector<bitset<200>> vb;
struct Maxclique {
    double limit=0.025, pk=0;
    struct Vertex { int i, d=0; };
    typedef vector<Vertex> vv;
    vb e;
    vv V;
    vector<vi> C;
    vi qmax, q, S, old;
    void init(vv& r) {
        for (auto& v : r) v.d = 0;
        for (auto& v : r) for (auto j : r) v.d += e[v.i][j.i];
        sort(all(r), [](auto a, auto b) { return a.d > b.d; });
        int mxD = r[0].d;
        rep(i, 0, sz(r)) r[i].d = min(i, mxD) + 1;
    }
    void expand(vv& R, int lev = 1) {
        S[lev] += S[lev-1] - old[lev];
        old[lev] = S[lev-1];
        while (sz(R)) {
            if (sz(q) + R.back().d <= sz(qmax)) return;
            q.push_back(R.back().i);
            vv T;
            for (auto v:R) if (e[R.back().i][v.i]) T.push_back({v.i});
            if (sz(T)) {
                if (S[lev]++ / ++pk < limit) init(T);
                int j = 0, mxk = 1, mnk = max(sz(qmax) - sz(q) + 1, 1);
                C[1].clear(), C[2].clear();
                for (auto v : T) {
                    int k = 1;
                    auto f = [&](int i) { return e[v.i][i]; };
                    while (any_of(all(C[k]), f)) k++;
                    if (k > mxk) mxk = k, C[mxk+1].clear();
                    if (k < mnk) T[j++].i = v.i;
                    C[k].push_back(v.i);
                }
                if (j > 0) T[j-1].d = 0;
                rep(k, mnk, mxk+1) for (int i : C[k])
                    T[j].i = i, T[j++].d = k;
                expand(T, lev+1);
            } else if (sz(q) > sz(qmax)) qmax = q;
            q.pop_back(), R.pop_back();
        }
    }
    vi maxClique() { init(V), expand(V); return qmax; }
    Maxclique(vb conn) : e(conn), C(sz(e)+1, S(sz(C)), old(S) {
        rep(i, 0, sz(e)) V.push_back({i});
    }
};
```

MaximumIndependentSet.h

Description: To obtain a maximum independent set of a graph, find a max clique of the complement. If the graph is bipartite, see MinimumVertexCover.

7.7 Trees

BinaryLifting.h

Description: Calculate power of two jumps in a tree, to support fast upward jumps and LCAs. Assumes the root node points to itself.

Time: construction $\mathcal{O}(N \log N)$, queries $\mathcal{O}(\log N)$

bfce85, 25 lines

```
vector<vi> treeJump(vi& P) {
    int on = 1, d = 1;
    while(on < sz(P)) on *= 2, d++;
    vector<vi> jmp(d, P);
    rep(i,1,d) rep(j,0,sz(P))
        jmp[i][j] = jmp[i-1][jmp[i-1][j]];
    return jmp;
}
```

```
int jmp(vector<vi>& tbl, int nod, int steps) {
    rep(i,0,sz(tbl))
        if(steps&(1<<i)) nod = tbl[i][nod];
    return nod;
}
```

```
int lca(vector<vi>& tbl, vi& depth, int a, int b) {
    if (depth[a] < depth[b]) swap(a, b);
    a = jmp(tbl, a, depth[a] - depth[b]);
    if (a == b) return a;
    for (int i = sz(tbl); i--;) {
        int c = tbl[i][a], d = tbl[i][b];
        if (c != d) a = c, b = d;
    }
    return tbl[0][a];
}
```

LCA.h

Description: Data structure for computing lowest common ancestors in a tree (with 0 as root). C should be an adjacency list of the tree, either directed or undirected.

Time: $\mathcal{O}(N \log N + Q)$

"../data-structures/RMQ.h"

0fb2fb, 21 lines

```
struct LCA {
    int T = 0;
    vi time, path, ret;
    RMQ<int> rmq;

    LCA(vector<vi>& C) : time(sz(C)), rmq((dfs(C,0,-1), ret)) {}
    void dfs(vector<vi>& C, int v, int par) {
        time[v] = T++;
        for (int y : C[v]) if (y != par) {
            path.push_back(v), ret.push_back(time[v]);
            dfs(C, y, v);
        }
    }

    int lca(int a, int b) {
        if (a == b) return a;
        tie(a, b) = minmax(time[a], time[b]);
        return path[rmq.query(a, b)];
    }
    //dist(a,b) {return depth[a] + depth[b] - 2*depth[lca(a,b)];}
};
```

CompressTree.h

Description: Given a rooted tree and a subset S of nodes, compute the minimal subtree that contains all the nodes by adding all (at most $|S| - 1$) pairwise LCA's and compressing edges. Returns a list of (par, orig_index) representing a tree rooted at 0. The root points to itself.

Time: $\mathcal{O}(|S| \log |S|)$

"LCA.h"

9775a0, 21 lines

```
typedef vector<pair<int, int>> vpi;
```

BinaryLifting LCA CompressTree HLD LinkCutTree

```
vpi compressTree(LCA& lca, const vi& subset) {
    static vi rev; rev.resize(sz(lca.time));
    vi li = subset, &T = lca.time;
    auto cmp = [&](int a, int b) { return T[a] < T[b]; };
    sort(all(li), cmp);
    int m = sz(li)-1;
    rep(i,0,m) {
        int a = li[i], b = li[i+1];
        li.push_back(lca.lca(a, b));
    }
    sort(all(li), cmp);
    li.erase(unique(all(li)), li.end());
    rep(i,0,sz(li)) rev[li[i]] = i;
    vpi ret = {pii(0, li[0])};
    rep(i,0,sz(li)-1) {
        int a = li[i], b = li[i+1];
        ret.emplace_back(rev[lca.lca(a, b)], b);
    }
    return ret;
}
```

HLD.h

Description: Decomposes a tree into vertex disjoint heavy paths and light edges such that the path from any leaf to the root contains at most $\log(n)$ light edges. Code does additive modifications and max queries, but can support commutative segtree modifications/queries on paths and subtrees. Takes as input the full adjacency list. VALS_EDGES being true means that values are stored in the edges, as opposed to the nodes. All values initialized to the segtree default. Root must be 0.

Time: $\mathcal{O}((\log N)^2)$

"../data-structures/LazySegmentTree.h"

9547af, 46 lines

```
template <bool VALS_EDGES> struct HLD {
    int N, tim = 0;
    vector<vi> adj;
    vi par, siz, rt, pos;
    Node *tree;
    HLD(vector<vi> adj_)
        : N(sz(adj_)), adj(adj_), par(N, -1), siz(N, 1),
          rt(N), pos(N), tree(new Node(0, N)) { dfsSz(0); dfsHld(0); }
    void dfsSz(int v) {
        for (int& u : adj[v]) {
            adj[u].erase(find(all(adj[u]), v));
            par[u] = v;
            dfsSz(u);
            siz[v] += siz[u];
            if (siz[u] > siz[adj[v][0]]) swap(u, adj[v][0]);
        }
    }
    void dfsHld(int v) {
        pos[v] = tim++;
        for (int u : adj[v]) {
            rt[u] = (u == adj[v][0] ? rt[v] : u);
            dfsHld(u);
        }
    }
    template <class B> void process(int u, int v, B op) {
        for (; v = par[rt[v]];) {
            if (pos[u] > pos[v]) swap(u, v);
            if (rt[u] == rt[v]) break;
            op(pos[rt[v]], pos[v] + 1);
        }
        op(pos[u] + VALS_EDGES, pos[v] + 1);
    }
    void modifyPath(int u, int v, int val) {
        process(u, v, [&](int l, int r) { tree->add(l, r, val); });
    }
    int queryPath(int u, int v) { // Modify depending on problem
        int res = -1e9;
        process(u, v, [&](int l, int r) {
```

```
            res = max(res, tree->query(l, r));
        });
        return res;
    }
    int querySubtree(int v) { // modifySubtree is similar
        return tree->query(pos[v] + VALS_EDGES, pos[v] + siz[v]);
    }
};
```

LinkCutTree.h

Description: Represents a forest of unrooted trees. You can add and remove edges (as long as the result is still a forest), and check whether two nodes are in the same tree.

Time: All operations take amortized $\mathcal{O}(\log N)$.

0fb462, 90 lines

```
struct Node { // Splay tree. Root's pp contains tree's parent.
    Node *p = 0, *pp = 0, *c[2];
    bool flip = 0;
    Node() { c[0] = c[1] = 0; fix(); }
    void fix() {
        if (c[0]) c[0]->p = this;
        if (c[1]) c[1]->p = this;
        // (+ update sum of subtree elements etc. if wanted)
    }
    void pushFlip() {
        if (!flip) return;
        flip = 0; swap(c[0], c[1]);
        if (c[0]) c[0]->flip ^= 1;
        if (c[1]) c[1]->flip ^= 1;
    }
    int up() { return p ? p->c[1] == this : -1; }
    void rot(int i, int b) {
        int h = i ^ b;
        Node *x = c[i], *y = b == 2 ? x : x->c[h], *z = b ? y : x;
        if ((y->p = p)) p->c[up()] = y;
        c[i] = z->c[i ^ 1];
        if (b < 2) {
            x->c[h] = y->c[h ^ 1];
            y->c[h ^ 1] = x;
        }
        z->c[i ^ 1] = this;
        fix(); x->fix(); y->fix();
        if (p) p->fix();
        swap(pp, y->pp);
    }
    void splay() {
        for (pushFlip(); p; ) {
            if (p->p) p->p->pushFlip();
            p->pushFlip(); pushFlip();
            int c1 = up(), c2 = p->up();
            if (c2 == -1) p->rot(c1, 2);
            else p->p->rot(c2, c1 != c2);
        }
    }
    Node* first() {
        pushFlip();
        return c[0] ? c[0]->first() : (splay(), this);
    }
};

struct LinkCut {
    vector<Node> node;
    LinkCut(int N) : node(N) {}

    void link(int u, int v) { // add an edge (u, v)
        assert(!connected(u, v));
        makeRoot(&node[u]);
        node[u].pp = &node[v];
    }
};
```



```

void cut(int u, int v) { // remove an edge (u, v)
    Node *x = &node[u], *top = &node[v];
    makeRoot(top); x->splay();
    assert(top == (x->pp ? x->c[0]));
    if (x->pp) x->pp = 0;
    else {
        x->c[0] = top->p = 0;
        x->fix();
    }
}

bool connected(int u, int v) { // are u, v in the same tree?
    Node* nu = access(&node[u])->first();
    return nu == access(&node[v])->first();
}

void makeRoot(Node* u) {
    access(u);
    u->splay();
    if(u->c[0]) {
        u->c[0]->p = 0;
        u->c[0]->flip ^= 1;
        u->c[0]->pp = u;
        u->c[0] = 0;
        u->fix();
    }
}

Node* access(Node* u) {
    u->splay();
    while (Node* pp = u->pp) {
        pp->splay(); u->pp = 0;
        if (pp->c[1]) {
            pp->c[1]->p = 0; pp->c[1]->pp = pp;
            pp->c[1] = u; pp->fix(); u = pp;
        }
    }
    return u;
}
};

```

DirectedMST.h

Description: Finds a minimum spanning tree/arborescence of a directed graph, given a root node. If no MST exists, returns -1.

Time: $\mathcal{O}(E \log V)$

".../data-structures/UnionFindRollback.h"

39e620, 60 lines

```

struct Edge { int a, b; ll w; };
struct Node {
    Edge key;
    Node *l, *r;
    ll delta;
    void prop() {
        key.w += delta;
        if (l) l->delta += delta;
        if (r) r->delta += delta;
        delta = 0;
    }
    Edge top() { prop(); return key; }
};

Node* merge(Node* a, Node* b) {
    if (!a || !b) return a ? b;
    a->prop(), b->prop();
    if (a->key.w > b->key.w) swap(a, b);
    swap(a->l, (a->r = merge(b, a->r)));
    return a;
}

void pop(Node*& a) { a->prop(); a = merge(a->l, a->r); }

```

```

pair<ll, vi> dmst(int n, int r, vector<Edge>& g) {
    RollbackUF uf(n);
    vector<Node*> heap(n);
    for (Edge e : g) heap[e.b] = merge(heap[e.b], new Node{e});
}

```

```

ll res = 0;
vi seen(n, -1), path(n), par(n);
seen[r] = r;
vector<Edge> Q(n), in(n, {-1,-1}), comp;
deque<tuple<int, int, vector<Edge>>> cycs;
rep(s, 0, n) {
    int u = s, qi = 0, w;
    while (seen[u] < 0) {
        if (!heap[u]) return {-1, {} };
        Edge e = heap[u]->top();
        heap[u]->delta -= e.w, pop(heap[u]);
        Q[qi] = e, path[qi++] = u, seen[u] = s;
        res += e.w, u = uf.find(e.a);
        if (seen[u] == s) {
            Node* cyc = 0;
            int end = qi, time = uf.time();
            do cyc = merge(cyc, heap[w = path[--qi]]);
            while (uf.join(u, w));
            u = uf.find(u), heap[u] = cyc, seen[u] = -1;
            cycs.push_front({u, time, {Q[qi], &Q[end]} });
        }
    }
    rep(i, 0, qi) in[uf.find(Q[i].b)] = Q[i];
}

for (auto& [u, t, comp] : cycs) { // restore sol (optional)
    uf.rollback(t);
    Edge inEdge = in[u];
    for (auto& e : comp) in[uf.find(e.b)] = e;
    in[uf.find(inEdge.b)] = inEdge;
}
rep(i, 0, n) par[i] = in[i].a;
return {res, par};
}

```

7.8 Math

7.8.1 Number of Spanning Trees

Create an $N \times N$ matrix **mat**, and for each edge $a \rightarrow b \in G$, do **mat[a][b]--**, **mat[b][b]++** (and **mat[b][a]--**, **mat[a][a]++** if G is undirected). Remove the i th row and column and take the determinant; this yields the number of directed spanning trees rooted at i (if G is undirected, remove any row/column).

7.8.2 Erdős–Gallai theorem

A simple graph with node degrees $d_1 \geq \dots \geq d_n$ exists iff $d_1 + \dots + d_n$ is even and for every $k = 1 \dots n$,

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k).$$

Geometry (8)

8.1 Geometric primitives

CW :: = ClockWise 顺时针

CCW :: = CounterClockWise 逆时针

Point.h

Description: 平面点类。T 为 double 或 long long 等类型 (避免 int).

constexpr double eps = 0;

```

template<class T> int sgn(T x) { return (x > eps)-(x < -eps); }
template<class T> struct Point {
    using P = Point;
    T x, y;
    explicit Point(T x = 0, T y = 0) : x(x), y(y) {}
    bool operator<(P p) const { return tie(x,y) < tie(p.x,p.y); }
    bool operator==(P p) const { return tie(x,y)==tie(p.x,p.y); }
    bool approxEq(P p) const { return !(sgn(x-p.x)|sgn(y-p.y)); }
    P operator+(P p) const { return P(x + p.x, y + p.y); }
    P operator-(P p) const { return P(x - p.x, y - p.y); }
    P operator*(T d) const { return P(x * d, y * d); }
    P operator/(T d) const { return P(x / d, y / d); }
    T dot(P p) const { return x * p.x + y * p.y; } // 点乘
    T cross(P p) const { return x * p.y - y * p.x; } // 叉乘
    // Oa与Ob叉乘
    T cross(P a, P b) const { return (a-*this).cross(b-*this); }
    T dist2() const { return x * x + y * y; } // 到原点距离平方
    double dist() const { return sqrt((double)dist2()); } // 到原点距离
    // 与x轴夹角, 范围[-pi, pi]
    double angle() const { return atan2(y, x); }
    P unit() const { return *this/dist(); } // 单位化
    P perp() const { return P(-y, x); } // 绕原点逆时针旋转九十度
    P normal() const { return perp().unit(); } // 法向量
    // 绕原点逆时针旋转a弧度
    P rotate(double a) const {
        return P(x*cos(a)-y*sin(a), x*sin(a)+y*cos(a));
    }
    friend ostream& operator<<(ostream& os, P p) {
        return os << "(" << p.x << ", " << p.y << ")";
    }
};

```

$$dis = \frac{|(e-s) \times (p-s)|}{|e-s|}$$

$$dis \leq eps \Leftrightarrow |(e-s) \times (p-s)| \leq |e-s| \cdot eps$$

sideOf.h

Description: 返回点 p 相对于线段 se 的位置。 $1/0/-1 \Leftrightarrow$ 左侧/在线上/右侧。如果给定了可选参数 eps , 则当点 p 距离线段 se 的距离小于 eps 时, 返回 0。P 应该是 $\text{Point}<\text{T}>$ 类型, 其中 T 可以是 double 或 long long 等类型。在中间步骤中使用了乘积运算, 因此如果使用 int 或 long long 类型, 请注意溢出。

Usage: bool left = sideOf(p1,p2,q)==1;

"Point.h"

3af81c, 9 lines

```

template<class P>
int sideOf(P s, P e, P p) { return sgn(s.cross(e, p)); }

```

```

template<class P>
int sideOf(const P& s, const P& e, const P& p, double eps) {
    auto a = (e-s).cross(p-s); // 避免 s==e
    double l = (e-s).dist()*eps;
    return (a > l) - (a < -l);
}

```

lineDistance.h

Description:

返回点 p 到包含点 a 和 b 的直线的有符号距离。从 a 到 b 看, 左侧为正值, 右侧为负值。a==b 时返回 nan。P 应该是 $\text{Point}<\text{T}>$ 或 $\text{Point3D}<\text{T}>$ 类型, 其中 T 可以是 double 或 long long 等类型。在中间步骤中使用了乘积运算, 因此如果使用 int 或 long long 类型, 请注意溢出。使用 Point3D 将始终给出非负距离。对于 Point3D , 请在叉积的结果上调用 .dist。

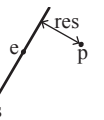
"Point.h"

f6bf6b, 4 lines

```

template<class P>
double lineDist(const P& a, const P& b, const P& p) {
    return (double)(b-a).cross(p-a)/(b-a).dist();
}

```



求直线交点

1/0/-1 $\Leftrightarrow$ 唯一交点/无交点/重合

$$L_1 : s_1 + t(e_1 - s_1)$$

$$L_2 : s_2 + u(e_2 - s_2)$$

$$\text{交点 } X = s_1 + t(e_1 - s_1) = s_2 + u(e_2 - s_2)$$

$$\overrightarrow{s_2 X} \parallel \overrightarrow{s_2 e_2} \Rightarrow (X - s_2) \times (e_2 - s_2) = 0$$

$$\Rightarrow (s_1 + t(e_1 - s_1) - s_2) \times (e_2 - s_2) = 0$$

$$\Rightarrow (s_1 - s_2) \times (e_2 - s_2) + t(e_1 - s_1) \times (e_2 - s_2) = 0$$

$$\Rightarrow t = \frac{(e_2 - s_2) \times (s_1 - s_2)}{(e_1 - s_1) \times (e_2 - s_2)}$$

$$X = s_1 + \frac{(e_2 - s_2) \times (s_1 - s_2)}{(e_1 - s_1) \times (e_2 - s_2)} \cdot (e_1 - s_1)$$

$$= \{[(e_1 - s_1) \times (e_2 - s_2) - (e_2 - s_2) \times (s_1 - s_2)] \cdot s_1 + [(e_2 - s_2) \times (s_1 - s_2)] \cdot e_1\} \cdot \frac{1}{(e_1 - s_1) \times (e_2 - s_2)}$$

$$= \{[(e_1 - s_1) \times (e_2 - s_2) + (s_1 - s_2) \times (e_2 - s_2)] \cdot s_1 + [(e_2 - s_2) \times (s_1 - s_2)] \cdot e_1\} \cdot \frac{1}{(e_1 - s_1) \times (e_2 - s_2)}$$

$$= \frac{[(e_1 - s_2) \times (e_2 - s_2)] \cdot s_1 + [(e_2 - s_2) \times (s_1 - s_2)] \cdot e_1}{(e_1 - s_1) \times (e_2 - s_2)}$$

$$= \frac{p \cdot s_1 + q \cdot e_1}{d}$$

lineIntersection.h

Description:

如果存在唯一的交点, 返回 {1, 交点}; 如果不存在交点, 返回 {0, (0,0)}; 如果存在无数个交点, 返回 {-1, (0,0)}。如果使用 Point<ll>, 且交点坐标不全为整数, 则返回错误的结果。中间步骤使用了三个坐标的乘积, 因此如果使用 int 或 ll, 请注意可能的溢出。

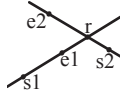
Usage: auto res = lineInter(s1,e1,s2,e2);

if (res.first == 1)

cout << "intersection point at " << res.second << endl;

"Point.h" a01f81, 8 lines

```
template<class P>
pair<int, P> lineInter(P s1, P e1, P s2, P e2) {
    auto d = (e1 - s1).cross(e2 - s2);
    if (d == 0) // if parallel
        return {-(s1.cross(e1, s2) == 0), P(0, 0)};
    auto p = s2.cross(e1, e2), q = s2.cross(e2, s1);
    return {1, (s1 * p + e1 * q) / d};
}
```



点线距离

$$q_{\leftrightarrow} = \text{proj}_{\overrightarrow{se}}(p)$$

$$(p - q_{\leftrightarrow}) \cdot (e - s) = 0,$$

$$q_{\leftrightarrow} = s + \lambda_{\leftrightarrow}(e - s),$$

$$(p - s - \lambda_{\leftrightarrow}(e - s)) \cdot (e - s) = 0,$$

$$(p - s) \cdot (e - s) = \lambda_{\leftrightarrow}((e - s) \cdot (e - s)),$$

$$\lambda_{\leftrightarrow} = \frac{(p - s) \cdot (e - s)}{\|e - s\|^2},$$

$$\lambda_{\overline{se}} = \text{clamp}\left(\frac{(p - s) \cdot (e - s)}{\|e - s\|^2}, 0, 1\right),$$

$$q_{\overline{se}} = s + \lambda_{\overline{se}}(e - s),$$

$$\text{segDist}(s, e, p) = \|p - q_{\overline{se}}\|.$$

SegmentDistance.h

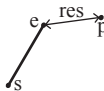
Description:

返回点 p 到线段 se 的最短距离。

Usage: Point<double> a, b(2,2), p(1,1);

bool onSegment = segDist(a,b,p) < 1e-10;

"Point.h"



cbce89, 7 lines

```
using P = Point<double>;
double segDist(P &s, P &e, P &p) {
    if (s == e) return (p - s).dist();
    auto d = (e-s).dist2(), t = max(.0, min(d, (p-s).dot(e-s)));
    // t = clamp((p - s).dot(e - s), 0, d);
    return ((p - s) * d - (e - s) * t).dist() / d;
}
```

点在线段上

$$(e - s) \times (p - s) = 0 \iff p \in \overleftrightarrow{se},$$

$$(p - s) \cdot (p - e) = 0 \iff p = s \text{ or } p = e,$$

$$(p - s) \cdot (p - e) < 0 \iff p \in \overline{se}.$$

OnSegment.h

Description: 当且仅当 p 位于线段 se 上时返回 true。使用 Point<double> 时改用 (segDist(s,e,p)<=epsilon)。

"Point.h", "SegmentDistance.h"

7fld77, 7 lines

```
template<class P> bool onSegment(P s, P e, P p) {
    return p.cross(s, e) == 0 && (s - p).dot(e - p) <= 0;
}
```

```
template<class P> bool onSegment(P s, P e, P p, double eps) {
    return segDist(s, e, p) <= eps;
}
```

线段交点

$$|res| = \begin{cases} 0, & \text{无交点,} \\ 1, & \text{唯一交点,} \\ 2, & \text{无限交点, 返回公共线段的两个端点.} \end{cases}$$

$$os_1 = \overrightarrow{s_2 e_2} \times \overrightarrow{s_2 s_1},$$

$$oe_1 = \overrightarrow{s_2 e_2} \times \overrightarrow{s_2 e_1},$$

$$os_2 = \overrightarrow{s_1 e_1} \times \overrightarrow{s_1 s_2},$$

$$oe_2 = \overrightarrow{s_1 e_1} \times \overrightarrow{s_1 e_2}.$$

$$\text{唯一交点 } p = s_1 + t(e_1 - s_1),$$

$$(e_2 - s_2) \times (p - s_2) = 0.$$

$$0 = (e_2 - s_2) \times (s_1 + t(e_1 - s_1) - s_2)$$

$$= (e_2 - s_2) \times (s_1 - s_2) + t(e_2 - s_2) \times (e_1 - s_1)$$

$$= (e_2 - s_2) \times (s_1 - s_2) + t(e_2 - s_2) \times ((e_1 - s_2) + (s_2 - s_1))$$

$$= os_1 + t(oe_1 - os_1).$$

$$t = \frac{os_1}{os_1 - oe_1}.$$

$$\begin{aligned} p &= s_1 + \frac{os_1}{os_1 - oe_1}(e_1 - s_1) \\ &= \frac{s_1(os_1 - oe_1) + os_1(e_1 - s_1)}{os_1 - oe_1} \\ &= \frac{s_1 oe_1 - e_1 os_1}{oe_1 - os_1}. \end{aligned}$$

SegmentIntersection.h

Description:

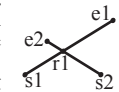
若从 s1 到 e1 的线段与从 s2 到 e2 的线段存在唯一交点, 则返回该交点。若不存在交点, 则返回空 vector。若存在无限多个交点, 则返回一个包含 2 个元素的 vector, 其中存放公共线段的两个端点。如果 P 是 Point<ll> 且交点坐标不是整数, 返回的位置会是错误的。中间步骤会用到三个坐标的乘积, 因此使用 int 或 long long 时要注意溢出。

Usage: vector<P> inter = segInter(s1,e1,s2,e2);

if (sz(inter)==1)

cout << "segments intersect at " << inter[0] << endl;

"Point.h", "OnSegment.h"



4ae7a2, 13 lines

```
template<class P> vector<P> segInter(P s1, P e1, P s2, P e2) {
    auto os1 = s2.cross(e2, s1), oe1 = s2.cross(e2, e1),
        os2 = s1.cross(e1, s2), oe2 = s1.cross(e1, e2);
    // 判断交点是否为唯一的非端点交点。
    if (sgn(os1) * sgn(oe1) < 0 && sgn(os2) * sgn(oe2) < 0)
        return {(s1 * oe1 - e1 * os1) / (oe1 - os1)};
    set<P> res;
    if (onSegment(s2, e2, s1)) res.insert(s1);
    if (onSegment(s2, e2, e1)) res.insert(e1);
    if (onSegment(s1, e1, s2)) res.insert(s2);
    if (onSegment(s1, e1, e2)) res.insert(e2);
    return {all(res)};
}
```

LineProjectionReflection.h

Description: 将点 p 投影到直线 ab 上。若设置 refl=true, 则改为返回点 p 关于直线 ab 的对称点。如果 P 是整数点且目标点坐标不是整数, 返回的点会是错误的。中间步骤会用到三个坐标的乘积, 因此要注意溢出。

"Point.h"

b5562d, 5 lines

```
template<class P>
P lineProj(P a, P b, P p, bool refl=false) {
    P v = b - a;
```

```
return p - v.perp()*(1+refl)*v.cross(p-a)/v.dist2();
}
```

linearTransformation.h

Description:

Apply the linear transformation (translation, rotation and scaling) which takes line p0-p1 to line q0-q1 to point r.

"Point.h" 03a306, 6 lines

```
typedef Point<double> P;
P linearTransformation(const P& p0, const P& p1,
    const P& q0, const P& q1, const P& r) {
    P dp = p1-p0, dq = q1-q0, num(dp.cross(dq), dp.dot(dq));
    return q0 + P((r-p0).cross(num), (r-p0).dot(num))/dp.dist2();
}
```

Strings (9)

KMP.h

Description: pi[x] 计算 s 的最长前缀长度, 且该前缀以 s[x] 结尾, 且不等于 s[0...x] 本身 (abacaba -> 0010123)。可以用来找到一个字符串的所有出现位置。

Time: $\mathcal{O}(n)$ d4375c, 16 lines

```
vi pi(const string& s) {
    vi p(sz(s));
    rep(i,1,sz(s)) {
        int g = p[i-1];
        while (g && s[i] != s[g]) g = p[g-1];
        p[i] = g + (s[i] == s[g]);
    }
    return p;
}
```

```
vi match(const string& s, const string& pat) {
    vi p = pi(pat + '\0' + s), res;
    rep(i,sz(p)-sz(s),sz(p))
        if (p[i] == sz(pat)) res.push_back(i - 2 * sz(pat));
    return res;
}
```

Zfunc.h

Description: 维护 [l, r] 表示当前已知最靠右的匹配区间 $s[l..r-1] == s[0..r-l-1]$ 。z[i] 计算 $s[i:]$ 与 s 的最长公共前缀, 但 $z[0] = 0$ 除外。(abacaba -> 0010301)

Time: $\mathcal{O}(n)$ ee09e2, 12 lines

```
vi Z(const string& S) {
    vi z(sz(S));
    int l = -1, r = -1;
    rep(i,1,sz(S)) {
        z[i] = i >= r ? 0 : min(r - i, z[i - l]);
        while (i + z[i] < sz(S) && S[i + z[i]] == S[z[i]])
            z[i]++;
        if (i + z[i] > r)
            l = i, r = i + z[i];
    }
    return z;
}
```

Manacher.h

Description: 对于字符串中的每个位置 i, $p[0][i]$ = 中心在 i-1 与 i 之间的最长偶回文半长, $p[1][i]$ = 中心在 i 的最长奇回文半长 (向下取整)(不包括 i)

Time: $\mathcal{O}(N)$ 09d7b8, 13 lines

```
array<vi, 2> manacher(const string& s) {
    int n = sz(s);
    array<vi, 2> p = {vi(n + 1, 0), vi(n, 0)};
    rep(id, 0, 2) for (int i = 0, l = 0, r = 0; i < n; i++) {
        int t = r - i + !id;
```

```
if (i < r) p[id][i] = min(t, p[id][l + t]);
int L = i - p[id][i], R = i + p[id][i] - !id;
while (L - 1 >= 0 && R + 1 <= n - 1 && s[L - 1] == s[R + 1])
    p[id][i]++, L--, R++;
if (R > r) l = L, r = R;
}
return p;
}
```

MinRotation.h

Description: 找到字符串中字典序最小的循环位移

Usage: rotate(v.begin(), v.begin()+minRotation(v), v.end());

Time: $\mathcal{O}(N)$ 7c0f7c, 8 lines

```
int minRotation(string s) {
    int a=0, n=sz(s); s += s;
    rep(b,0,n) rep(k,0,n) {
        if (a+k == b || s[a+k] < s[b+k]) { b += max(0, k-1); break; }
        if (s[a+k] > s[b+k]) { a = b; break; }
    }
    return a;
}
```

AC 自动机

Aho-Corasick 自动机用于 多模式串匹配

给定若干模式串:

he
she
his
hers

再给一个文本串:ushers

要找出每个模式串在哪些位置出现。朴素做法是每个模式串跑一次 KMP, 总复杂度约为: $\mathcal{O}(\text{模式串个数} \times \text{文本长度})$

AC 自动机的目标:

构建: $\mathcal{O}(\text{所有模式串总长} \times \text{字符集大小})$

find: $\mathcal{O}(\text{文本长度} + \text{匹配输出量})$

findAll: 最坏达 $\mathcal{O}(\text{文本长度} \times \text{模式串数})$

直接输出链: $\text{end} \rightarrow \dots \rightarrow \text{start}$,

fail 输出链: $N[y].\text{end} \rightarrow \dots$,

拼接后: $\text{end} \rightarrow \dots \rightarrow \text{start} \rightarrow N[y].\text{end} \rightarrow \dots$.

AhoCorasick.h

Description: Aho-Corasic 自动机, 用于多模式串匹配。用 AhoCorasick ac(patterns) 初始化; 自动机的起始节点下标为 0。find(word) 对每个位置返回在该位置结束的最长模式串编号; 若不存在则返回 -1。findAll(-, word) 找出从每个位置开始的所有模式串 (若无重复模式串, 数量最多为 $N\sqrt{N}$), 按长度从短到长排列。允许重复模式串; 不允许空模式串。若要找出从每个位置开始的最长模式串, 反转所有输入即可。对大字符集, 可将每个符号拆成若干块, 并用哨兵位标记符号边界。

Time: construction takes $\mathcal{O}(26N)$, where N = sum of length of patterns. find(x) is $\mathcal{O}(N)$, where N = length of x. findAll is $\mathcal{O}(NM)$. 91f62d, 73 lines

```
vi encode(vi a) {
    vi v;
    for (int i = 0; i < sz(a); v.push_back(256), i++)
```

```
rep(i, 0, 4)v.push_back((a[i] >> (8 * i)) & 256);
return v;
}
```

```
struct AhoCorasick {
    enum { alpha = 26, first = 'A' }; // change this!
    struct Node {
        // (nmatches is optional)
        int back = 0, next[alpha], start = -1, end = -1, nmatches=0;
        Node(int v) { memset(next, v, sizeof(next)); }
    };
    vector<Node> N;
    vi backp;
    void insert(string& s, int j) {
        assert(!s.empty());
        int n = 0;
        for (char c : s) {
            int& m = N[n].next[c - first];
            if (m == -1) { n = m = sz(N); N.emplace_back(-1); }
            else n = m;
        }
        if (N[n].end == -1) N[n].start = j;
        backp.push_back(N[n].end);
        N[n].end = j;
        N[n].nmatches++;
    }
    AhoCorasick(vector<string>& pat) : N(1, -1) {
        rep(i,0,sz(pat)) insert(pat[i], i);
        N[0].back = sz(N);
        N.emplace_back(0);

        queue<int> q;
        for (q.push(0); !q.empty(); q.pop()) {
            int n = q.front(), prev = N[n].back;
            rep(i,0,alpha) {
                int& ed = N[n].next[i], y = N[prev].next[i];
                if (ed == -1) ed = y;
                else {
                    N[ed].back = y;
                    (N[ed].end == -1 ? N[ed].end : backp[N[ed].start])
                        = N[y].end;
                    N[ed].nmatches += N[y].nmatches;
                    q.push(ed);
                }
            }
        }
    }
    vi find(string word) {
        int n = 0;
        vi res; // ll count = 0;
        for (char c : word) {
            n = N[n].next[c - first];
            res.push_back(N[n].end);
            // count += N[n].nmatches;
        }
        return res;
    }
    vector<vi> findAll(vector<string>& pat, string word) {
        vi r = find(word);
        vector<vi> res(sz(word));
        rep(i,0,sz(word)) {
            int ind = r[i];
            while (ind != -1) {
                res[i - sz(pat[ind]) + 1].push_back(ind);
                ind = backp[ind];
            }
        }
        return res;
    }
}
```

```
};
```

后缀数组

定义. 将字符串的所有后缀按字典序排序, 只保存后缀起点:

$\text{sa}[i]$ = 字典序第 i 小的后缀起点.

若 $s=\text{banana}$, 排序后为

$a, \text{ana}, \text{anana}, \text{banana}, \text{na}, \text{nana},$

因此 $\text{sa} = [5, 3, 1, 0, 4, 2]$. 反数组通常记为 rk :

$\text{rk}[\text{pos}]$ = 后缀 $s[\text{pos} \dots]$ 的排名.

本板子构造结束后用 $\text{x}[i]$ 承担 $\text{rk}[i]$ 的角色.

哨兵. 在 s 后加入严格小于所有字符的哨兵。本板子使用

$\text{s.push_back}(0)$

故输入不能包含 `nul` 字符。若原串长为 n , 则哨兵所在的空后缀最小:

$\text{sa}[0] = n.$

所以返回数组大小为 $n+1$, 处理原串后缀时常从 $\text{sa}[1]$ 开始看。

循环位移到后缀数组. `cp-algorithms` 的构造本质上先排序 **循环位移**而非直接排序后缀。循环位移把字符串看成首尾相接, 位置 i 开始的循环串为

$s[i \dots] + s[0 \dots i-1].$

更一般地, 当区间右端绕回开头时:

$s[i..j] = s[i..n-1] + s[0..j] \quad (i > j),$

下标按模 n 理解。

排序循环位移如何变成排序后缀? 在原串后加入最小哨兵后, 任何真实后缀都可看作一个循环位移中哨兵之前的前缀。由于哨兵严格小于所有普通字符, 两个后缀第一次分出长短时, 较短者会先遇到哨兵, 字典序也正好更小。因此

sort cyclic shifts of $(s + '0')$ $\iff$ sort suffixes of s ,

只不过结果最前面多出哨兵位置对应的空后缀。本板子不删除它, 所以保留 $\text{sa}[0] = n$ 。

例如 $\text{dabbb\$}$ 的循环位移排序中, 去掉哨兵后的前缀顺序为

$\text{abbb}, \text{b}, \text{bb}, \text{bbb}, \text{dabbb},$

这正是 dabbb 的后缀字典序。

倍增思想. 已知每个位置开始、长度为 j 的子串排名后, 长度为 $2j$ 的子串可由二元组表示:

$(\text{rank}[i], \text{rank}[i+j]).$

本板子中

$\text{sa} = p,$ $\text{x} = c,$ y = 临时起点/旧排名, ws = 计数桶,

其中 p, c 对应 `cp-algorithms` 中的有序位置数组与等价类数组。主循环

$j = 0, 1, 2, 4, 8, \dots$

每轮把已知长度翻倍。第一次 $j=0$ 可视为按单字符排序, 因为初始化时已有 $\text{x}[i]=s[i]$ 。

SuffixArray LIS



基数排序. 每轮需要按

$(\text{x}[i], \text{x}[i+j])$

排序。直接比较二元组会多一个 $\log n$, 因此使用计数排序。代码先构造 y , 使其已按第二关键字有序; 再稳定地按第一关键字 $\text{x}[\text{y}[i]]$ 计数排序:

$\text{sa}[\text{--ws}[\text{x}[\text{y}[i]]]] = \text{y}[i].$

于是每轮为 $O(n)$, 轮数为 $O(\log n)$, 总复杂度

$O(n \log n).$

重编号. 排序完成后, 相邻后缀 $\text{a}=\text{sa}[i-1], \text{b}=\text{sa}[i]$ 的新排名由旧二元组决定:

$(\text{old}[\text{a}], \text{old}[\text{a}+j]) = (\text{old}[\text{b}], \text{old}[\text{b}+j]) \Rightarrow \text{new}[\text{a}] = \text{new}[\text{b}].$

对应代码为

$(\text{y}[\text{a}] == \text{y}[\text{b}] \ \&\& \ \text{y}[\text{a}+j] == \text{y}[\text{b}+j]).$

若不同排名数 p 已达到 n , 所有后缀排名唯一, 构造结束。

LCP/height. 后缀数组常配合相邻最长公共前缀数组:

$\text{lcp}[i] = \text{LCP}(s[\text{sa}[i] \dots], s[\text{sa}[i-1] \dots]), \quad \text{lcp}[0] = 0.$

构造完 sa 后, $\text{x}[i]$ 即 $\text{rk}[i]$ 。后缀 i 在排序中的前一个后缀为

$j = \text{sa}[\text{x}[i]-1].$

Kasai 思想: 若上一次公共前缀长度为 k , 起点右移一位后至少可从 $k-1$ 继续尝试, 因此代码中有

$k \ \&\& \ k--.$

LCP 构造总复杂度为 $O(n)$ 。

常用公式. 任意两个后缀在 sa 中排名为 $l < r$ 时:

$\text{LCP}(\text{sa}[l], \text{sa}[r]) = \min_{i=l+1}^r \text{lcp}[i].$

不同子串个数:

$$\frac{n(n+1)}{2} - \sum_i \text{lcp}[i].$$

最长重复子串:

$$\max_i \text{lcp}[i].$$

至少出现 k 次的最长子串:

$$\max_l \min_{i=l+1}^{l+k-1} \text{lcp}[i].$$

两个串最长公共子串: 构造

$$S = A + \# + B,$$

枚举相邻后缀, 若来自不同原串, 用 $\text{lcp}[i]$ 更新答案。

SuffixArray.h

Description: 构造字符串的后缀数组。 $\text{sa}[i]$ 表示后缀数组中第 i 小后缀的起始位置。返回数组大小为 $n+1$, 且 $\text{sa}[0] = n$ 。lcp 数组保存后缀数组中相邻后缀的最长公共前缀: $\text{lvp}[i] = \text{lcp}(\text{sa}[i], \text{sa}[i-1]), \text{lcp}[0] = 0$ 。输入字符串不能包含空字符。

Time: $O(n \log n)$

635552, 22 lines

```
struct SuffixArray {
    vi sa, lcp;
    SuffixArray(string s, int lim=256) { // or vector<int>
        s.push_back(0); int n = sz(s), k = 0, a, b;
        vi x(all(s)), y(n), ws(max(n, lim));
        sa = lcp = y, iota(all(sa), 0);
        for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p) {
            p = j, iota(all(y), n - j);
            rep(i, 0, n) if (sa[i] >= j) y[p++] = sa[i] - j;
            fill(all(ws), 0);
            rep(i, 0, n) ws[x[i]]++;
            rep(i, 1, lim) ws[i] += ws[i - 1];
            for (int i = n; i--;) sa[--ws[x[i]]] = y[i];
            swap(x, y), p = 1, x[sa[0]] = 0;
            rep(i, 1, n) a = sa[i - 1], b = sa[i], x[b] =
                (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++;
        }
        for (int i = 0, j; i < n - 1; lcp[x[i++]] = k)
            for (k && k--, j = sa[x[i] - 1];
                s[i + k] == s[j + k]; k++);
    }
};
```

Various (10)

10.1 Misc. algorithms

LIS.h

Description: 返回最长上升 (非降) 子序列的下标。

Time: $O(N \log N)$

277a90, 17 lines

```
template<class I> vi lis(const vector<I> &S) {
    if (S.empty()) return {};
    vi prev(sz(S));
    using p = pair<I, int>;
    vector<p> res;
    rep(i, 0, sz(S)) {
        // 把lower_bound中的 0 -> i 以求最长非降子序列
        auto it = lower_bound(all(res), p{S[i], 0});
        if (it == res.end()) res.emplace_back(), it = res.end()-1;
        *it = {S[i], i};
        prev[i] = it == res.begin() ? 0 : (it - 1)->second;
    }
    int L = sz(res), cur = res.back().second;
    vi ans(L);
    while (L--) ans[L] = cur, cur = prev[cur];
    return ans;
}
```